Specialized Lending with Limited Pledgeability

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Abstract

We develop a model of credit cycles in which specialized lenders can preserve borrower going-concern value after foreclosure, allowing firms to borrow against value that traditional collateral-based lenders cannot fully recover. Limited pledgeability arises because this recovery advantage cannot be fully transferred to the intermediary's own creditors, so specialized lending requires intermediary balance-sheet resources. With heterogeneous borrowers, specialized intermediaries expand credit per unit of collateral, especially to firms with less transferable, more intangible value. Following sufficiently large adverse shocks, however, scarce intermediary resources induce a *flight-to-pledgeability*, shifting new lending toward borrowers with more transferable collateral. We then apply the model to the private credit market and show that the presence of specialized intermediaries matches several established facts about lending cycles and private credit specialization. We also test and find supportive evidence for flight-to-pledgeability, using Business Development Company lending during COVID-19 as our laboratory.

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1 Introduction

Over the last two decades, private credit has expanded rapidly as an alternative to bank lending and public debt markets. Despite charging higher interest rates,[1] specialized lenders can expand firms' borrowing capacity by preserving going-concern value that traditional collateral-based lenders cannot fully recover. However, this recovery advantage cannot be fully pledged to the specialized lenders' own creditors, making credit provision dependent on intermediary balance-sheet resources. This creates a fundamental trade-off: specialized lending expands access to credit but exposes borrowers to fluctuations in intermediary financial constraints. How does this trade-off shape the aggregate supply and allocation of credit over the business cycle?

We study this question in a financial accelerator model in which specialized lenders have an advantage in recovering borrower going-concern value. Three main findings emerge. First, specialized lending expands debt capacity, particularly for firms with less transferable, more intangible value, but its long-run effects on investment and production depend on borrowers' wealth accumulation. Second, specialized intermediaries cushion the initial contraction in credit following adverse shocks, but the subsequent recovery is slower. Third, when intermediary resources become sufficiently scarce, lending shifts away from borrowers with less pledgeable assets, generating a *flight-to-pledgeability*. We provide empirical evidence consistent with this prediction using business development company lending during the COVID-19 crisis.

The key friction is that specialized lenders can recover going-concern value that they cannot fully pledge to their own creditors. The model contains two forms of collateral value, a general and a specialized one. The general value is transferable: it can be realized by essentially any creditor through liquidation or resale. The specialized value instead can be realized only by a lender with the relevant expertise or operational capacity to keep the borrower's business operating. A specialized lender may therefore recover substantially more from a distressed borrower than an outside financier. Crucially, however, the specialized lender cannot pledge the entire recovery value to its own creditors. It can pledge only the

[1] Chernenko et al. (2022) and Haque et al. (forthcoming) show that the premium over bank loans persists after controlling for borrower and loan characteristics, while Cai and Haque (2024) documents higher average spreads relative to institutional Term Loan B loans. IMF (2024) documents higher borrowing costs in private credit relative to syndicated leveraged loans and public corporate bonds.

transferable component[2], as outside lenders can recover only the market value of the assets in case of liquidation. Specialized lending therefore creates additional borrowing capacity for firms, but doing so requires the specialized intermediary to commit its own balance-sheet resources.

We introduce this feature into a financial accelerator model in the spirit of Kiyotaki and Moore (1997), where investment is lumpy. Following their agrarian metaphor, we refer to borrowers as farmers and outside financiers as gatherers, and introduce *gardeners*, specialized intermediaries that can preserve the value of seized projects after foreclosure. Farmers invest by planting trees on their land, but these trees are lost if the land is transferred to gatherers. Gardeners can instead preserve the trees and their continuation value. Farmers can therefore borrow against this additional value from gardeners, who in turn can pledge only the market value of bare land to gatherers. The difference between these values depends on equilibrium market conditions, while the additional lending requires gardener equity, making specialized credit dependent on intermediary balance sheets.

Our model first clarifies when specialization matters. Suppose a seized project could change hands without loss. Its competitive market price already incorporates the value generated by its most productive use. An outside financier holding it could then sell it at the market price, and so would recover exactly what the specialist would have recovered by operating it; borrowers would obtain the same credit from either lender. A productive advantage in operating collateral expands debt capacity only when the project cannot be transferred without destroying value, so that the specialist recovers more by keeping it than an outside financier recovers by selling it. This makes non-transferability, rather than lender productivity by itself, central to the mechanism.

When such non-transferable value is present, specialized lending changes borrowing and investment even in normal times. Because a specialized lender can recover more from a project in foreclosure, a borrower can credibly promise it a larger repayment; the same project therefore supports more debt, and the borrower needs a smaller equity contribution when new investment opportunities arise. The immediate effect of specialized-lender entry is therefore an expansion in borrowing and investment. The same loans, however, commit more of the project's future payoff to creditors, so the borrower reaches its next investment opportunity with less internal wealth. As highly leveraged borrower vintages accumulate, the

[2]This idea relates to inalienability of human capital by Hart and Moore (1994).

initial investment expansion is gradually reversed. Appendix A.3.6 illustrates this transition in the calibrated economy. In the model's steady state, specialized lending raises debt capacity and leverage per unit of wealth but reduces borrower land holdings, production, and equilibrium asset prices. The effect of specialized credit is therefore fundamentally dynamic: easier financing today changes borrower wealth accumulation tomorrow.

We next study how specialized lending affects the transmission of adverse shocks. Following moderate shocks, specialized intermediaries absorb losses by reducing payouts, allowing them to continue lending against borrowers' going-concern value. Once payouts are exhausted, however, intermediary resources become scarce and lending against non-transferable value becomes increasingly costly. The response of credit to shocks is therefore nonlinear, with credit conditions tightening more sharply as intermediary balance-sheet constraints bind. Nevertheless, in our numerical exercises, specialized lending attenuates the initial contraction relative to an economy with traditional collateral-based lending, even following large shocks. This resilience comes at the cost of a slower recovery. Because a specialized lender recovers value by operating the project over its remaining life, the borrower's debt capacity depends on expected land prices over that horizon. After an adverse shock, land prices are expected to recover only gradually; debt capacity therefore falls by less than the price of land, which limits the initial reduction in borrowers' land holdings. Because borrowers make only a small down payment on each unit of land, however, a given loss of wealth requires a large total reduction in their land holdings over time. Limiting the initial reduction therefore defers the remainder and prolongs the downturn. Specialized lending thus dampens the immediate effects of adverse shocks but increases their persistence.

We then extend the model to borrowers that differ in pledgeability. Some projects depend heavily on value that only the specialized lender can preserve, as with intangible assets, while others are backed by collateral that can more readily be transferred to outside financiers, as with tangible assets. In normal times, specialized lending is especially valuable to the first group because it allows those firms to borrow against a larger share of their continuation value. The same dependence makes these borrowers the most exposed when the intermediary comes under stress. Loans to low-pledgeability borrowers tie up more of the specialized lender's internal resources, so when those resources become scarce, debt capacity and new originations fall more sharply for these borrowers and lending shifts toward those whose collateral outside financiers can fund. This is the flight to pledgeability. Because both

groups are financed from the same balance sheet, the reallocation does not depend on where losses originate: a sufficiently large loss in one part of the lender's portfolio reduces its capacity to finance borrowers elsewhere, including firms whose own fundamentals have not deteriorated. A common intermediary balance sheet creates cross-sector contagion.

Finally, we test the flight-to-pledgeability prediction using business development companies (BDCs), specialized non-bank lenders to middle-market firms, around the COVID-19 shock. BDCs provide an empirical setting to test the model's predictions: they lend against continuation value and carry part of those loans on their own balance sheets. In the model, a shock to the lender's internal resources reduces each borrower's debt capacity in proportion to its pledgeability wedge; a shock to the lender's portfolio should therefore hit less pledgeable borrowers harder. We measure the lender shock by a predetermined exposure to COVID-19, derived from the industry composition of each BDC's portfolio at the end of 2019, and we measure borrower pledgeability by the industry liquidation recovery rates of Kermani and Ma (2023), with higher recovery corresponding to a smaller wedge in the model. Using quarterly PitchBook data on BDC–borrower relationships, we estimate, across BDCs lending to the same borrower, how the cumulative growth of credit from 2019Q4 to a given endpoint varies with lender exposure, and how that relation depends on the recovery rate of the borrower's industry. In the spirit of Khwaja and Mian (2008), the borrower fixed effects absorb the borrower's own demand and performance, so identification comes from differences across lenders within a borrower. Among BDCs lending to the same firm, the more exposed lender cuts credit more, and this gap is largest for borrowers in low-recovery industries: at the end of 2021, a BDC ten percentage points more exposed has a 26-percentage-point more favorable cumulative credit growth for a borrower one standard deviation higher in industry recovery. The interaction exhibits no pre-trend before the shock and fades by 2024.

Literature and context. On the theoretical side, our paper contributes to the literature on financial amplification through borrower and intermediary balance sheets. In Bernanke et al. (1999) and Kiyotaki and Moore (1997), shocks to borrower net worth and collateral values tighten borrowing constraints and amplify fluctuations. Shleifer and Vishny (1992) emphasize the role of liquidation values in determining debt capacity, while Rampini and Viswanathan (2010) study the allocation of scarce debt capacity under collateral constraints. A complementary literature places intermediary balance sheets at the center of financial

amplification (Holmstrom and Tirole, 1997; He and Krishnamurthy, 2013; Brunnermeier and Sannikov, 2014). We connect these mechanisms by showing how specialized lenders' ability to preserve borrower going-concern value expands debt capacity but makes credit provision dependent on intermediary balance-sheet resources. This dependence generates nonlinear responses to adverse shocks: credit tightens more sharply once intermediary resources become scarce.

Our closest theoretical antecedent is Rampini and Viswanathan (2019). In both models, intermediaries can support borrowing against value that cannot be fully pledged to outside financiers, requiring intermediary capital to sustain additional debt capacity. Our contribution is to endogenize this pledgeability wedge through creditor-specific recovery values, rather than assuming a fixed difference in collateral enforcement ability. Specialized lenders can preserve going-concern value following foreclosure that would otherwise be lost upon transfer to outside financiers. Because this recovery advantage depends on equilibrium market conditions and borrower characteristics, the intermediary-capital requirement varies across assets and states. This generates a new credit allocation mechanism: when intermediary resources become scarce, lending shifts toward borrowers whose collateral value is more transferable. Moreover, because borrower losses endogenously erode intermediary resources, financial distress can propagate across borrowers through a common intermediary balance sheet, even when their own fundamentals remain unchanged.

Our paper also contributes to the growing literature on private credit by providing a unified theoretical framework for several documented facts. First, private lenders and private equity sponsors frequently maintain borrower relationships through distress rather than immediately liquidating their claims (Jang, 2024). Second, private credit can stabilize lending over the credit cycle (Hinzen et al., 2026), potentially at the cost of slower recoveries associated with prolonged distress resolution (Rintamäki and Steffen, 2025). Third, direct lenders disproportionately finance intangible-intensive firms (Haque et al., 2022) and invest in monitoring and industry expertise (Jang et al., 2025; Block et al., 2024). Fourth, private equity sponsors and direct lenders' expertise supports higher borrower leverage (Banegas et al., 2026; Chernenko et al., 2022). Fifth, we document that credit agreements between banks and business development companies (BDCs) assign lower advance rates and higher haircuts to portfolio assets that are harder for outside financiers to operate or value, including junior, non-tradable, and non-performing loans. This provides direct evidence that specialized

intermediaries' assets differ in their pledgeability to outside creditors. Our model rationalizes these facts through a common mechanism: specialized intermediaries create value by preserving and financing borrower continuation value that traditional creditors cannot fully recover. Beyond rationalizing existing evidence, the model generates a new prediction of flight-to-pledgeability during severe downturns, which we test using BDC lending during COVID-19. This evidence contributes to the empirical literature on BDCs and the transmission of intermediary financial constraints (Davydiuk et al., 2024b,c,a; Chernenko et al., 2025).

Finally, our paper relates to the corporate finance literature distinguishing asset-based from cash-flow-based lending (Benmelech et al., 2026; Lian and Ma, 2021; Ivashina et al., 2022; Kermani and Ma, 2023). While traditional collateral-based lending relies on assets that can be transferred and liquidated, cash-flow-based lending may depend more heavily on preserving borrowers' continuation value. Our model provides a theoretical foundation for this distinction by showing that financing non-transferable value requires specialized recovery capabilities and intermediary balance-sheet resources. This mechanism is particularly relevant as the composition of corporate assets and debt financing evolves, potentially creating opportunities for specialized lenders to finance firms with limited tangible collateral.

Roadmap. The paper proceeds as follows. Section 2 develops the model. Section 3 relates the model to recent evidence on private credit and private equity. Section 4 tests the model's reallocation prediction using BDC lending around COVID-19, and Section 5 concludes.

2 Model

We develop a credit-cycle model in the spirit of Kiyotaki and Moore (1997) to study the role of specialized lenders in the provision of credit. The model introduces a simple form of creditor-specific enforcement. Specialized lenders can preserve part of a borrower's going-concern value following foreclosure, whereas outside lenders can recover only the transferable value of collateral. This recovery advantage expands debt capacity, but requires specialized lenders to commit their own resources. As these resources become scarce, borrowing capacity contracts and shocks can propagate through the lender balance sheet.

2.1 Setting

2.1.1 Environment

Time is discrete, $t = 0, 1, 2, \ldots$. There is a perishable tradable fruit, which is the numeraire, and a fixed aggregate supply K of land. Land trades competitively at price q_t. Land can be bare or cultivated. Cultivated land consists of the underlying land and a stock of trees, which survive from one period to the next at rate $\lambda \in (0, 1)$.

There are three groups of agents. A unit mass of farmers operates cultivated land and borrows against it. A mass $n > 0$ of identical gardeners can hold land and lend to farmers. We interpret gardeners as specialized lenders. A mass $m > 0$ of identical gatherers can hold land, lend directly to farmers against bare-land collateral, and provide funding to gardeners. Gatherers represent outside financiers who cannot operate the investment.

Let x_t^F, x_t^S, and x_t^G denote consumption by farmers, gardeners, and gatherers. Preferences are linear:

$$U^F = \sum_{t=0}^{\infty} \beta^t x_t^F, \tag{1}$$

$$U^S = \sum_{t=0}^{\infty} (\beta^S)^t x_t^S, \tag{2}$$

$$U^G = \sum_{t=0}^{\infty} (\beta')^t x_t^G, \tag{3}$$

where

$$0 < \beta < \beta^S \leq \beta' < 1. \tag{4}$$

Thus, farmers are more impatient than gardeners, which are weakly more impatient than gatherers.

There is a competitive one-period credit market. Gatherers can lend freely in this market, so their Euler equation pins down the safe gross interest rate:

$$R \equiv \frac{1}{\beta'} \tag{5}$$

Debt is expressed in principal units: principal B_t issued at gross rate R_t at date t requires

repayment $R_t B_t$ at date $t+1$.

Within each date, output from assets held at the end of the previous date is realized first and outstanding debt becomes due. Farmers then observe their investment opportunities and choose land holdings, new one-period debt, and consumption. They repay inherited debt using harvest revenue, land sales, and new borrowing. As in Kiyotaki and Moore (1997), farmers supply the labor required for the current harvest before an unanticipated productivity shock. They therefore cannot renegotiate inherited debt by threatening to withdraw that labor. Future labor remains uncommitted, so the human-capital friction continues to limit new and refinanced debt.[3]

2.1.2 Production and land

Let k_t^F denote cultivated land held by an individual farmer at the end of date t, and let

$$K_t^F \equiv \mathbb{E}[k_t^F] \tag{6}$$

denote aggregate farmer land. Since farmers have unit mass, K_t^F is also average farmer land. Each unit produces $a > 0$ units of tradable fruit and $c > 0$ units of nontradable "bruised fruit." Aggregate tradable output and farmer consumption are therefore

$$Y_{t+1}^F = aK_t^F, \tag{7}$$

$$x_{t+1}^F = cK_t^F. \tag{8}$$

Bruised fruit cannot be traded or pledged.

Let k_t^S and k_t^G denote land held by an individual gardener and gatherer. Their tradable production technologies are $H(k_t^S)$ and $G(k_t^G)$, respectively, with

$$H' > 0, \qquad H'' < 0, \qquad G' > 0, \qquad G'' < 0. \tag{9}$$

Aggregate gardener and gatherer land holdings are nk_t^S and mk_t^G. Land-market clearing requires

$$K_t^F + nk_t^S + mk_t^G = K. \tag{10}$$

[3]See Kiyotaki and Moore (1997), p. 224, footnote 13.

Gatherer land demand satisfies

$$q_t - \frac{q_{t+1}}{R} = \frac{1}{R} G'(k_t^G), \tag{11}$$

for an interior landholding. Thus, the standard user cost of land is

$$u_t \equiv q_t - \frac{q_{t+1}}{R} \tag{12}$$

2.1.3 Lumpy investment

As in Section III of Kiyotaki and Moore (1997) - KM97 hereinafter, investment is lumpy. At each date, a farmer receives an investment opportunity with probability $\pi \in (0, 1)$. Only farmers receiving an opportunity can plant new trees. Planting one unit costs $\phi > 0$ units of tradable fruit.

A farmer without an investment opportunity cannot replace trees that have died. Its noninvestment policy is to retain surviving trees, so it holds

$$k_t^F = \lambda k_{t-1}^F \tag{13}$$

A farmer with an investment opportunity can replace depreciated trees and expand cultivated land. If it ends date t with k_t^F units of cultivated land, investment expenditure is

$$\phi \left(k_t^F - \lambda k_{t-1}^F \right) \tag{14}$$

The stock of trees is specific to the farmer's project. A gardener that takes over the project can continue operating the surviving trees. If cultivated land is instead transferred to a gatherer, the trees are lost and the gatherer receives bare land. Farmers may also sell land to meet inherited obligations, destroying any surviving trees and receiving only its bare-land value, without recovering past planting expenditure. Because investment opportunities are lumpy, these trees cannot necessarily be recreated immediately. This assumption generates non-transferable going-concern value.

Let b_t^F denote the debt position of an individual farmer at the end of date t, with a positive value denoting borrowing and a negative value denoting saving. Aggregate net farmer debt is $B_t^F = \mathbb{E}[b_t^F]$, including borrowing from both lenders and netting out farmer savings.

Each borrowing farmer issues a single one-period claim secured by its land. It borrows directly from a gatherer at R if its entire loan is at most $q_{t+1}k_t^F/R$. Otherwise, a gardener supplies the entire loan and holds its foreclosure rights. Farmers can change lenders freely at renewal and place savings with gatherers.[4]

Let R_t^F be the gross contractual rate on the individual farmer's loan. For $b_t^F > 0$, competitive pricing gives

$$R_t^F = R + \left(\frac{1}{s_t} - R\right) \max\left\{1 - \frac{q_{t+1}k_t^F}{Rb_t^F}, 0\right\}, \tag{15}$$

where s_t is the gardener's shadow discount factor defined in (35). For saving and zero positions, we have $R_t^F = R$. Parameters are such that farmers receiving an investment opportunity find it optimal to invest up to the collateral constraint. At the steady state, borrowing up to the collateral constraint pays

$$R^F = \frac{L}{q/R + \beta^S(L-q)}$$

Inherited debt is due at the rate agreed at origination, to the lender that issued the loan, even after an unexpected change in land prices. The farmer budget is therefore:

$$a_t k_{t-1}^F + b_t^F = q_t(k_t^F - k_{t-1}^F) + \phi(k_t^F - \lambda k_{t-1}^F) + R_{t-1}^F b_{t-1}^F \tag{16}$$

Current harvest and new borrowing finance land purchases, planting, and repayment of inherited debt. A farmer without an investment opportunity retains its surviving trees, $k_t^F = \lambda k_{t-1}^F$, and sells the land whose trees have died. Its new debt or saving position is therefore

$$b_t^F = R_{t-1}^F b_{t-1}^F - \left[a_t + (1-\lambda)q_t\right] k_{t-1}^F. \tag{17}$$

A positive position is financed by gatherers if fully covered by bare land and by gardeners otherwise.

[4]Fully covered loans could instead remain with gardeners, who would then refinance them entirely by borrowing from gatherer. With competitive pricing, free refinancing and full repayment, this leaves aggregate allocations unchanged; see Appendix A.2.1.

2.2 Specialized lending

The distinction between gardeners and gatherers comes from their recovery values. Let L_{t+1} denote the maximum tradable value that a gardener can recover at date $t+1$ from one unit of cultivated farmer land. Farmer debt is subject to

$$R_t^F b_t^F \leq L_{t+1} k_t^F \tag{18}$$

This constraint binds for farmers receiving an investment opportunity.

Define

$$\chi_t \equiv \frac{L_{t+1}}{R_t^F} \tag{19}$$

as the maximum principal that an investing farmer can borrow per unit of cultivated land with R_t^F evaluated at the binding collateral constraint. Combining that constraint with (16) gives

$$k_t^F = \frac{(a_t + q_t + \lambda\phi)k_{t-1}^F - R_{t-1}^F b_{t-1}^F}{q_t + \phi - \chi_t} \tag{20}$$

The farmer's required equity contribution per unit of cultivated land is therefore

$$\text{downpayment} \equiv q_t + \phi - \chi_t. \tag{21}$$

Let $B_t^{F,S}$ denote farmer loans held by gardeners and $V_{t+1}^{F,S}$ their transferable value at repayment, anticipated at origination. The difference $B_t^{F,S} - V_{t+1}^{F,S}/R$ is financed with gardener resources. Aggregate farmer repayment, net of returns on farmer savings, is

$$\mathbb{E}[R_t^F b_t^F] = RB_t^F + \left(\frac{1}{s_t} - R\right)\left(B_t^{F,S} - \frac{V_{t+1}^{F,S}}{R}\right). \tag{22}$$

Since investment opportunities are iid, aggregation of the individual investment policy gives

$$K_t^F = \underbrace{(1-\pi)\lambda K_{t-1}^F}_{\substack{\text{Land retained by} \\ \text{non-investing farmers}}} + \underbrace{\frac{\pi}{q_t + \phi - \chi_t}}_{\substack{\text{Investing share} \\ / \text{ down payment}}} \left[\begin{array}{l} \underbrace{(a_t + q_t + \lambda\phi)K_{t-1}^F}_{\text{Assets and harvest}} - \underbrace{RB_{t-1}^F}_{\text{Repayment at } R} \\ - \underbrace{\left(\frac{1}{s_{t-1}} - R\right)\left(B_{t-1}^{F,S} - \frac{V_t^{F,S}}{R}\right)}_{\text{Gardener interest premium}} \end{array} \right] \tag{23}$$

$$B_t^F = \underbrace{RB_{t-1}^F}_{\text{Repayment at } R} + \underbrace{\left(\frac{1}{s_{t-1}} - R\right)\left(B_{t-1}^{F,S} - \frac{V_t^{F,S}}{R}\right)}_{\text{Gardener interest premium}} + \underbrace{q_t(K_t^F - K_{t-1}^F)}_{\text{Net land purchases}} + \underbrace{\phi(K_t^F - \lambda K_{t-1}^F)}_{\text{Planting expenditure}} - \underbrace{a_t K_{t-1}^F}_{\text{Harvest income}} . \tag{24}$$

Importantly, the collateral constraint binds for investing farmers, not for the farmer sector as a whole. As such,

$$\mathbb{E}\left[R_t^F b_t^F\right] \leq L_{t+1} K_t^F \tag{25}$$

with strict inequality in general when $\pi < 1$.

2.2.1 Foreclosure value

A gatherer that forecloses on farmer collateral can recover only the bare-land value q_{t+1}. A gardener can instead operate the surviving trees and liquidate the underlying land as trees die.

A gardener that retains the surviving trees can borrow against their underlying bare land. Retaining λ units therefore supports borrowing of $\lambda q_{t+2}/R$ at date $t+1$, with repayment λq_{t+2} at date $t+2$. Under perfect foresight, the value at date $t+1$ of operating one unit of

foreclosed cultivated land is

$$\widehat{L}_{t+1} = (1-\lambda)q_{t+1} + \lambda\left[\frac{q_{t+2}}{R} + s_{t+1}\left(a_{t+2} + L_{t+2} - q_{t+2}\right)\right]. \tag{26}$$

Here $s_{t+1} = \Lambda_{t+2}/\Lambda_{t+1}$ is the gardener's shadow discount factor, defined in (35). Borrowing provides funds at once and is repaid from next period's harvest and continuation value. The value L_{t+2} includes the option to liquidate next period.

The gardener chooses between operating and immediately liquidating the collateral:

$$L_{t+1} = \max\left\{q_{t+1}, \widehat{L}_{t+1}\right\} \tag{27}$$

Define the pledgeability wedge as

$$W_{t+1} \equiv L_{t+1} - q_{t+1} \geq 0. \tag{28}$$

When $W_{t+1} > 0$, the gardener can preserve value that is lost if collateral passes to a gatherer. We refer to this case as the *operational foreclosure regime*.

In a deterministic steady state with positive gardener consumption, $s = \beta^S$ and

$$L = q + \frac{\lambda \max\{\beta^S a - (1-1/R)q, 0\}}{1-\beta^S\lambda}. \tag{29}$$

Hence, we are in the operational foreclosure regime when

$$\beta^S a > \left(1 - \frac{1}{R}\right) q. \tag{30}$$

The inequality has a simple interpretation. Gardeners preserve the trees when their future harvest is sufficiently valuable relative to bare land.

2.2.2 Gardener balance sheet and loan pricing

Gardeners finance their farmer loan book $B_t^{F,S}$ with their own resources and one-period borrowing from gatherers. Let B_t^S denote aggregate gardener borrowing. Their budget

constraint is

$$\underbrace{nq_t(k_t^S - k_{t-1}^S)}_{\text{Net land purchases}} + \underbrace{B_t^{F,S}}_{\text{Farmer lending}} + \underbrace{RB_{t-1}^S}_{\text{Repayment to gatherers}} + \underbrace{x_t^S}_{\text{Consumption}}$$

$$= \underbrace{nH(k_{t-1}^S)}_{\text{Own production}} + \underbrace{V_t^{F,S} + \frac{B_{t-1}^{F,S} - V_t^{F,S}/R}{s_{t-1}}}_{\text{Farmer loan repayments}} + \underbrace{B_t^S}_{\text{Borrowing from gatherers}} \tag{31}$$

Gardeners use resources to purchase land, make farmer loans, repay gatherers, and consume. Their resources come from their own production, repayment of existing farmer loans, and new gatherer borrowing.

If gardeners default, gatherers seize gardener land and the farmer loan book. Gatherers can liquidate gardener land at its market value, but cannot preserve the gardener-specific continuation value embedded in farmer collateral. Their recovery from farmer loans is therefore limited to the bare-land value of the collateral. Summing these collateral values across loans held by gardeners gives $V_{t+1}^{F,S}$. Gardener borrowing is consequently subject to

$$RB_t^S \leq nq_{t+1}k_t^S + V_{t+1}^{F,S}. \tag{32}$$

We focus on the region in which this constraint binds. Appendix A.2 describes the vintage structure used to compute $V_{t+1}^{F,S}$.

Let Λ_t denote the marginal value of one additional unit of tradable resources inside the gardener sector, and let $\xi_t \geq 0$ denote the multiplier on $x_t^S \geq 0$. The gardener consumption condition is

$$(\beta^S)^t - \Lambda_t + \xi_t = 0, \qquad \xi_t x_t^S = 0, \tag{33}$$

so

$$\Lambda_t = (\beta^S)^t + \xi_t. \tag{34}$$

Define the shadow discount factor

$$s_t \equiv \frac{\Lambda_{t+1}}{\Lambda_t} = \frac{(\beta^S)^{t+1} + \xi_{t+1}}{(\beta^S)^t + \xi_t}. \tag{35}$$

When gardener consumption is positive at both dates, $\xi_t = \xi_{t+1} = 0$ and hence $s_t = \beta^S$.

Let $\mu_t \geq 0$ denote the multiplier on the gardener borrowing constraint (32). The optimality condition for gardener borrowing is

$$\Lambda_t = R\Lambda_{t+1} + R\mu_t. \tag{36}$$

Dividing by Λ_t and using the definition of s_t gives

$$\frac{\mu_t}{\Lambda_t} = \frac{1}{R} - s_t. \tag{37}$$

Gatherers supply fully covered farmer loans at R. Otherwise, the gardener advances b_t^F, borrowing $q_{t+1}k_t^F/R$ against the bare-land claim and providing the remainder with its own resources. It receives $R_t^F b_t^F$ next period and repays $q_{t+1}k_t^F$ to gatherers. Competitive pricing gives

$$b_t^F - \frac{q_{t+1}k_t^F}{R} = s_t \left(R_t^F b_t^F - q_{t+1}k_t^F\right). \tag{38}$$

Solving for the rate gives (15). The contractual rate is a weighted average of R and $1/s_t$, with the weight on R equal to the share of the loan financed against bare land. As this share approaches one, the contractual rate approaches R. Additional borrowing against the same land must be financed entirely with gardener resources, so each extra dollar borrowed adds $1/s_t$ dollars to next period's repayment.

For newly issued farmer loans at the collateral limit, the optimality condition is

$$\Lambda_t = R_t^F \Lambda_{t+1} + \mu_t \frac{q_{t+1}R_t^F}{L_{t+1}}. \tag{39}$$

The last term reflects the additional gatherer funding that one unit of farmer lending can support. A newly originated farmer loan at the collateral limit satisfies $R_t^F b_t^F = L_{t+1}k_t^F$, while gatherers can recover only $q_{t+1}k_t^F$. Hence one additional unit of farmer loan principal increases the pledgeable value of the gardener's loan book by $q_{t+1}R_t^F/L_{t+1}$.

Dividing (39) by $\Lambda_t R_t^F$ and using (37) yields

$$\frac{1}{R_t^F} = s_t + \left(\frac{1}{R} - s_t\right)\frac{q_{t+1}}{L_{t+1}}. \tag{40}$$

Proposition 1 (Gardener debt capacity)**.** *Suppose gardener consumption is positive at both*

dates, so that $s_t = \beta^S$. The maximum debt capacity of the farmer is

$$\chi_t = \frac{q_{t+1}}{R} + \beta^S \left(L_{t+1} - q_{t+1}\right) \tag{41}$$

Debt capacity consists of two components. The first, q_{t+1}/R, is borrowing backed by collateral that can be transferred to gatherers. The second, $\beta^S(L_{t+1} - q_{t+1})$, is the additional borrowing capacity generated by the gardener's recovery advantage. Because this recovery advantage is not pledgeable to gatherers, the second component must be financed by gardener resources.

Fully transferable benchmark. If productive assets are fully transferable, a gardener's foreclosure value equals the competitive land price:

$$L_{t+1} = q_{t+1} \tag{42}$$

Equations (40) and (41) then imply

$$R_t^F = R, \qquad \chi_t = \frac{q_{t+1}}{R} \tag{43}$$

The model therefore collapses to the KM97 financing constraint when the gardener has no creditor-specific recovery advantage. Appendix A provides the formal derivation.

Lemma 1 (Neutrality of gardener lending)**.** *When land is fully transferable and no productive capital is lost when ownership changes, gardener lending does not increase farmer debt capacity. Farmers can borrow q_{t+1}/R per unit of land whether they borrow from gardeners or directly from gatherers.*

2.3 Steady state allocation

In an initial steady state without gardener lending, farmers borrow from gatherers and can pledge only the transferable value of their land. Their debt capacity is therefore q^0/R per unit of cultivated land. Gardeners can increase farmer debt capacity only if they value seized cultivated land above its liquidation value.

At the initial steady state, gardener consumption is positive. Evaluating equation (29)

at the initial land price gives

$$L^0 = q^0 + \frac{\lambda \max\{\beta^S a - (1 - 1/R)q^0, 0\}}{1 - \beta^S \lambda}. \tag{44}$$

Lemma 2 (Gardener entry)**.** *Gardeners have a strict recovery advantage at the initial steady state if and only if*

$$\beta^S a > \left(1 - \frac{1}{R}\right) q^0. \tag{45}$$

Gardeners preserve the trees when the discounted harvest exceeds the cost of retaining the land. By retaining land worth q^0, they forgo its sale proceeds but can borrow q^0/R against it. A higher harvest or a lower land price therefore makes continued operation more attractive.

We first characterize the deterministic steady state in the operational foreclosure regime. When gardener consumption is positive, $s = \beta^S$, the farmer's debt capacity is

$$\chi = \frac{q}{R} + \beta^S (L - q). \tag{46}$$

Lemma 3 (Borrower leverage under gardener finance)**.** *Consider an operational steady state with positive gardener consumption. At a given land price q, an investing farmer financed by gardeners has higher debt relative to output than a farmer financed directly by gatherers:*

$$\frac{\chi}{a} - \frac{q}{aR} = \frac{\beta^S (L - q)}{a} > 0. \tag{47}$$

Gardeners can lend against the continuation value $L - q$ that is lost when collateral passes to a gatherer. They therefore allow the farmers they finance to borrow more per unit of operating income at the prevailing land price. Stationarity of farmer land requires

$$K^F = (1 - \pi)\lambda K^F + \pi \frac{(a + q + \lambda\phi)K^F - RB^F - \left(\frac{1}{\beta^S} - R\right)\left(B^{F,S} - \frac{V^{F,S}}{R}\right)}{q + \phi - \chi} \tag{48}$$

The first term is cultivated land carried by farmers without an investment opportunity. The second is the land held after investment by the fraction of farmers able to replant.

The aggregate farmer budget gives

$$(R-1)B^F + \left(\frac{1}{\beta^S} - R\right)\left(B^{F,S} - \frac{V^{F,S}}{R}\right) = \left[a - (1-\lambda)\phi\right] K^F. \tag{49}$$

The right-hand side is farmer output net of the cost of replacing dead trees. In steady state, this surplus pays net interest on farmer debt: interest at the safe rate, plus the premium on the part of lending financed with gardener resources. Hence,

$$\frac{B^F}{K^F} = \frac{a-(1-\lambda)\phi}{R-1} - \frac{1/\beta^S - R}{R-1}\frac{B^{F,S} - V^{F,S}/R}{K^F}. \tag{50}$$

Since aggregate farmer output is aK^F, net debt relative to output is

$$\frac{B^F}{aK^F} = \frac{a-(1-\lambda)\phi}{a(R-1)} - \frac{1/\beta^S - R}{a(R-1)}\frac{B^{F,S} - V^{F,S}/R}{K^F}. \tag{51}$$

For newly investing farmers, equation (40) implies

$$\frac{1}{R^F} = \beta^S + (\beta' - \beta^S)\frac{q}{L} \leq \beta' = \frac{1}{R}, \tag{52}$$

with strict inequality when $\beta^S < \beta'$. Since $B^{F,S} - V^{F,S}/R \geq 0$ and $1/\beta^S \geq R$, equation (51) gives

$$\frac{B^F}{aK^F} \leq \frac{a-(1-\lambda)\phi}{a(R-1)}, \tag{53}$$

with equality when $\beta^S = \beta'$ and strict inequality when $\beta^S < \beta'$ and $B^{F,S} > V^{F,S}/R$.

The right-hand side is the net debt-to-output ratio under direct gatherer finance. The premium on gardener funding reduces the net debt that a given operating surplus can support.

Steady-state gardener and gatherer land holdings satisfy

$$q\left(1 - \frac{1}{R}\right) = \beta^S H'(k^S), \tag{54}$$

$$q\left(1 - \frac{1}{R}\right) = \frac{1}{R}G'(k^G), \tag{55}$$

together with

$$K^F + nk^S + mk^G = K \tag{56}$$

These equations, together with (29), (40), and the gardener balance sheet, characterize the steady-state equilibrium.

Between investment opportunities, farmers sell land whose trees have died and use the income from these sales and their harvests to repay debt. In steady state, their debt per unit of remaining land declines.

Lemma 4 (Borrower age and source of finance)**.** *In a steady state with $L > q$ and positive gardener consumption, let $W = L - q$. Farmers whose last investment occurred j periods ago borrow from gardeners if and only if*

$$(\lambda\beta^S)^j > \frac{a}{a+W}.$$

For these farmers, aggregate debt $B^{F,j}$ and land holdings $K^{F,j}$ satisfy

$$\frac{B^{F,j}}{K^{F,j}} = \frac{q}{R} + \beta^S \left[W + a - a(\lambda\beta^S)^{-j}\right] > \frac{q}{R}.$$

Once the repayment on any remaining loan is fully covered by the bare-land value of their collateral, farmers borrow from gatherers at R. If they receive no further investment opportunity, they eventually repay their debt and begin saving with gatherers.

Farmers that get an investment opportunity borrow χ per unit of land from gardeners. Appendix A.2 proves the lemma and derives aggregate borrowing and saving.

Gardener lending has two opposing effects on farmers. At a given land price, it increases debt capacity and reduces the own funds needed to acquire cultivated land. The larger loan, however, raises future repayments and leaves the farmer with fewer resources at its next investment opportunity. This effect on farmer wealth persists after the loan is refinanced with gatherers. Proposition 8 in Appendix A.3 formalizes this comparison.

Compare steady states with and without gardener lending, keeping populations, technologies, and investment opportunities fixed. In both economies, farmer land is positive, and gardeners and gatherers have positive consumption and interior landholdings.

Proposition 2 (Steady-state effect of gardener lending)**.** *If $L > q$ in the economy with gardener lending, gardener lending lowers the stationary land price q, farmer land K^F, farmer consumption x^F, and farmer production Y^F. At a given land price, it raises farmer debt capacity and leverage at origination.*

In steady state, the reduction in resources available for subsequent investment dominates the initial increase in borrowing capacity. Appendix A.3 proves this result, and Appendix A.3.6 shows that when gardeners enter an unintermediated economy, credit and investment first expand before the decline in farmer wealth reverses the expansion.

2.4 Credit cycles

We now study the response to an unanticipated one-period decline in farmer productivity. The economy starts from the deterministic steady state in the operational foreclosure regime, so $L^* > q^*$ and $R^{F*} > R$. At date 0,

$$a_0 = (1-\Delta)a, \qquad a_t = a \quad \text{for all } t \geq 1, \qquad \Delta \in (0,1). \tag{57}$$

All end-of-date -1 stocks are at their steady-state values.

After date 0, productivity returns immediately to a, but the economy does not return immediately to steady state. The shock changes farmer and gardener balance sheets, and the resulting state variables determine the subsequent path of land prices, borrowing, and investment.

The aggregate farmer equations away from steady state are

$$K_t^F = (1-\pi)\lambda K_{t-1}^F + \pi \frac{(a_t + q_t + \lambda\phi)K_{t-1}^F - RB_{t-1}^F - \left(\frac{1}{s_{t-1}} - R\right)\left(B_{t-1}^{F,S} - \frac{V_t^{F,S}}{R}\right)}{q_t + \phi - \chi_t}, \tag{58}$$

and

$$B_t^F = RB_{t-1}^F + \left(\frac{1}{s_{t-1}} - R\right)\left(B_{t-1}^{F,S} - \frac{V_t^{F,S}}{R}\right) + q_t(K_t^F - K_{t-1}^F) + \phi(K_t^F - \lambda K_{t-1}^F) - a_t K_{t-1}^F \tag{59}$$

Around the steady state, the impact response of farmer land satisfies

$$\left.\frac{dK_0^F}{d\Delta}\right|_{\Delta=0} = \frac{K^{F*}}{q^* + \phi - \chi^*}\Bigg[-(1-\pi)(1-\lambda)\left.\frac{dq_0}{d\Delta}\right|_{\Delta=0} + [1-(1-\pi)\lambda]\left.\frac{d\chi_0}{d\Delta}\right|_{\Delta=0} - \pi a\Bigg] \tag{60}$$

The shock therefore affects farmer investment directly through current cash flow and indirectly through equilibrium debt capacity.

2.4.1 A small temporary shock

Debt capacity can be written as

$$\chi_t = \frac{q_{t+1}}{R} + s_t(L_{t+1} - q_{t+1}). \tag{61}$$

For comparison, in the KM97 economy,

$$\chi_t^{KM} = \frac{q_{t+1}}{R} \tag{62}$$

The credit-capacity gap is therefore

$$\chi_t - \chi_t^{KM} = s_t(L_{t+1} - q_{t+1}) \geq 0. \tag{63}$$

For sufficiently small shocks, gardener consumption remains positive at dates 0 and 1, so $s_0 = \beta^S$. Differentiating debt capacity with respect to the shock gives

$$\frac{d\chi_0}{d\Delta} = \frac{1}{R}\frac{dq_1}{d\Delta} + \beta^S\left(\frac{dL_1}{d\Delta} - \frac{dq_1}{d\Delta}\right). \tag{64}$$

In the KM97 benchmark,

$$\frac{d\chi_0^{KM}}{d\Delta} = \frac{1}{R}\frac{dq_1}{d\Delta} \tag{65}$$

The recovery value L_1 depends on the expected path of future land prices and harvests,

whereas gatherer recovery depends only on q_1. When

$$\frac{dL_1}{d\Delta} \geq \frac{dq_1}{d\Delta}, \tag{66}$$

gardener recovery falls weakly less than bare-land value, cushioning the decline in farmer debt capacity.

Proposition 3 (Attenuation for small shocks)**.** *Suppose the economy starts in the operational foreclosure regime, gardener consumption remains positive at dates* 0 *and* 1*, and condition* (66) *holds. For a given land-price path, farmer debt capacity contracts weakly less after a small adverse productivity shock under gardener finance than under KM97 financing.*

The attenuation comes from two sources. First, farmers begin with greater debt capacity because $L > q$. Second, the gardener recovery value depends on the continuation value of the project and is therefore less sensitive to a temporary decline in the current liquidation value of land.

Figure 2 illustrates this case for a small shock. While most of the difference reflects the larger share of land held by gatherers in the gardener economy, the figure shows differences in dynamics that are in part due to the going-concern value of land that can now be recovered. Farmer land contracts less on impact than in the KM97 economy but returns to steady state much more slowly. Along the recovery, user costs remain below their steady-state value, so W_t rises and debt capacity falls by less than bare-land value, which lets farmers keep more land early on. Farmers rebuild their net worth from the excess return on land (net of the cost of debt, relative to the steady state), which is positive only while land is cheap to hold. Because they release less land, its user cost falls by less: land prices climb back only slowly, so each period's capital gain adds little to farmers' net worth and debt capacity, and land holdings take longer to recover.

2.4.2 Buffer exhaustion

The gardener buffer is finite. Define

$$\Delta^A \equiv \inf\left\{\Delta > 0 : x_0^S(\Delta) = 0\right\} \tag{67}$$

While gardener consumption remains positive at dates 0 and 1, $s_0 = \beta^S$.

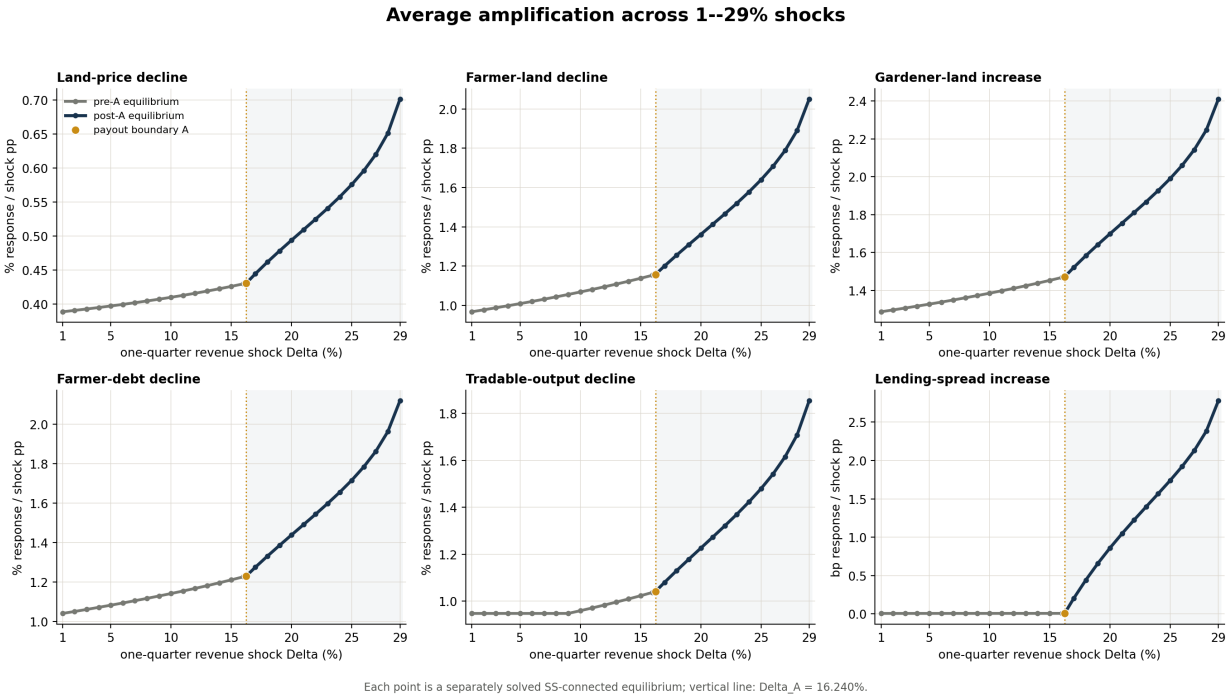


Figure 1: Peak response around gardener consumption buffer exhaustion

Notes: Peak responses per percentage point of the one-period productivity shock. The vertical line denotes the shock Δ^A at which gardener consumption reaches zero.

The consumption KKT condition implies

$$s_t = \frac{(\beta^S)^{t+1} + \xi_{t+1}}{(\beta^S)^t + \xi_t} \tag{68}$$

If current gardener resources are scarce, $\xi_0 > 0$, while consumption at date 1 remains positive, then $\xi_1 = 0$ and

$$s_0 = \frac{\beta^S}{1 + \xi_0} < \beta^S. \tag{69}$$

As ξ_0 rises, gardeners require a higher return on their funds and lend less against the non-transferable component of recovery.

Allowing s_0 to vary, the response of debt capacity becomes

$$\frac{d\chi_0}{d\Delta} = \frac{1}{R}\frac{dq_1}{d\Delta} + s_0\left(\frac{dL_1}{d\Delta} - \frac{dq_1}{d\Delta}\right) + (L_1 - q_1)\frac{ds_0}{d\Delta} \tag{70}$$

The final term is absent below Δ^A. Therefore, when s_0 falls, the final term adds to the contraction in debt capacity.

Because

$$R_0^F = \frac{L_1}{\chi_0}, \tag{71}$$

holding L_1 and q_1 fixed, a fall in s_0 raises the borrowing rate for investing farmers:

$$\frac{\partial R_0^F}{\partial s_0} < 0. \tag{72}$$

Proposition 4 (Buffer exhaustion and nonlinear amplification)**.** *For shocks below Δ^A, gardener consumption adjusts and the shadow discount factor remains at β^S. Once gardener consumption is exhausted, further shocks that increase the marginal value of current gardener resources lower s_0. In the operational foreclosure regime, a decline in s_0 reduces farmer debt capacity by $(L_1 - q_1)|ds_0|$ and raises the farmer borrowing rate. The response of credit therefore becomes more sensitive to additional losses after Δ^A.*

Figure 1 shows the nonlinear response.[5] In the numerical economy, gardener consumption is exhausted at a shock of approximately 16%. Below this threshold, gardener consumption adjusts to absorb losses. Above it, the marginal value of gardener resources rises and debt capacity becomes more sensitive to the shock as gardener equity is consumed.

Figure 3 shows the same comparison for a 19% shock, above the buffer-exhaustion threshold. The gardener economy is now more sensitive to additional losses than it was below Δ^A, but its initial contraction remains smaller than in the KM97 economy. The KM97 economy contracts more sharply and recovers quickly, whereas the gardener economy contracts less on impact and remains below steady state for longer.

2.4.3 Collapse of the economy

Once the consumption buffer is exhausted, the gardener needs to deplete its own resources to fund loans, up to the point the gardening sector cannot afford lending. We can define the

[5] In these counterfactual exercises, we assume that inherited non-investing vintages can refinance their debt as long as investing vintages can.

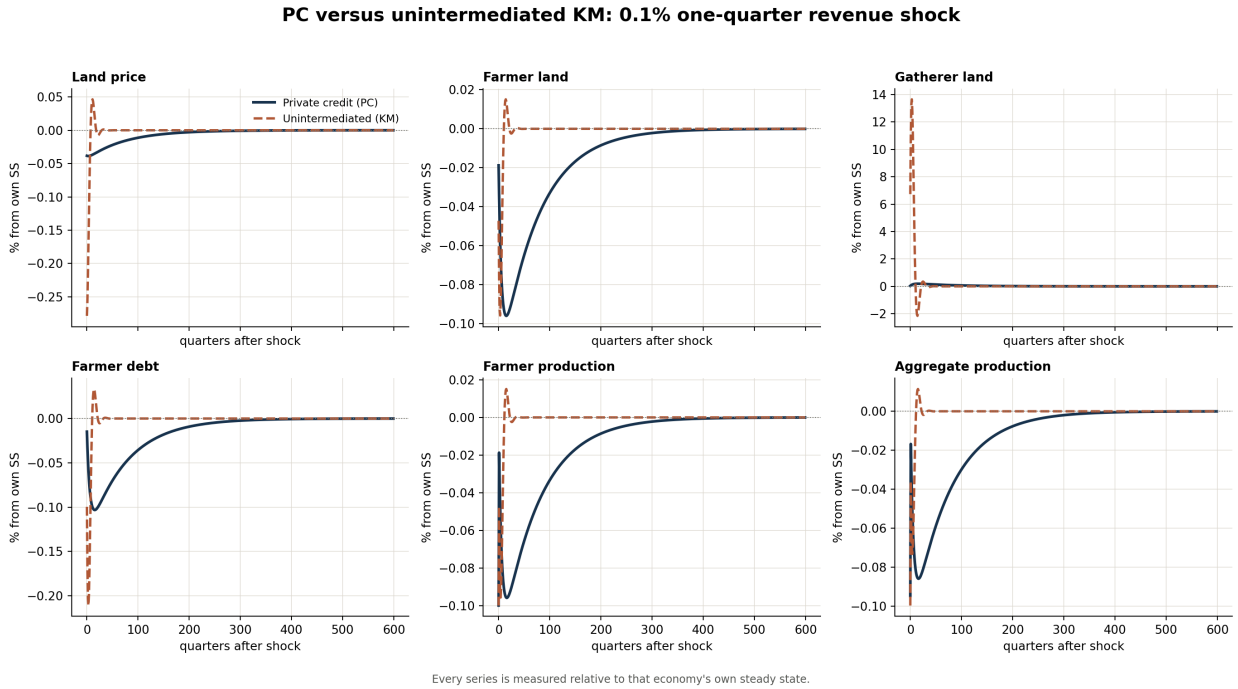


Figure 2: A small temporary shock: gardener lending versus KM

Notes: Responses to a 0.1% one-period decline in farmer tradable harvest, relative to each economy's own steady state. The solid line is the economy with gardener lending; the dashed line is the corresponding KM economy.

collapse threshold as

$$\Delta^C \equiv \inf\{\Delta > 0 : \text{gardeners can no longer meet their inherited obligations}\} \tag{73}$$

Define gardener net worth as

$$N_t^S \equiv nq_t k_t^S + B_t^{F,S} - B_t^S. \tag{74}$$

When the gardener borrowing constraint binds, combining (32) with (74) gives

$$N_t^S = n\left(q_t - \frac{q_{t+1}}{R}\right)k_t^S + B_t^{F,S} - \frac{V_{t+1}^{F,S}}{R} \tag{75}$$

The first term is the gardener resources required to support its own land after borrowing

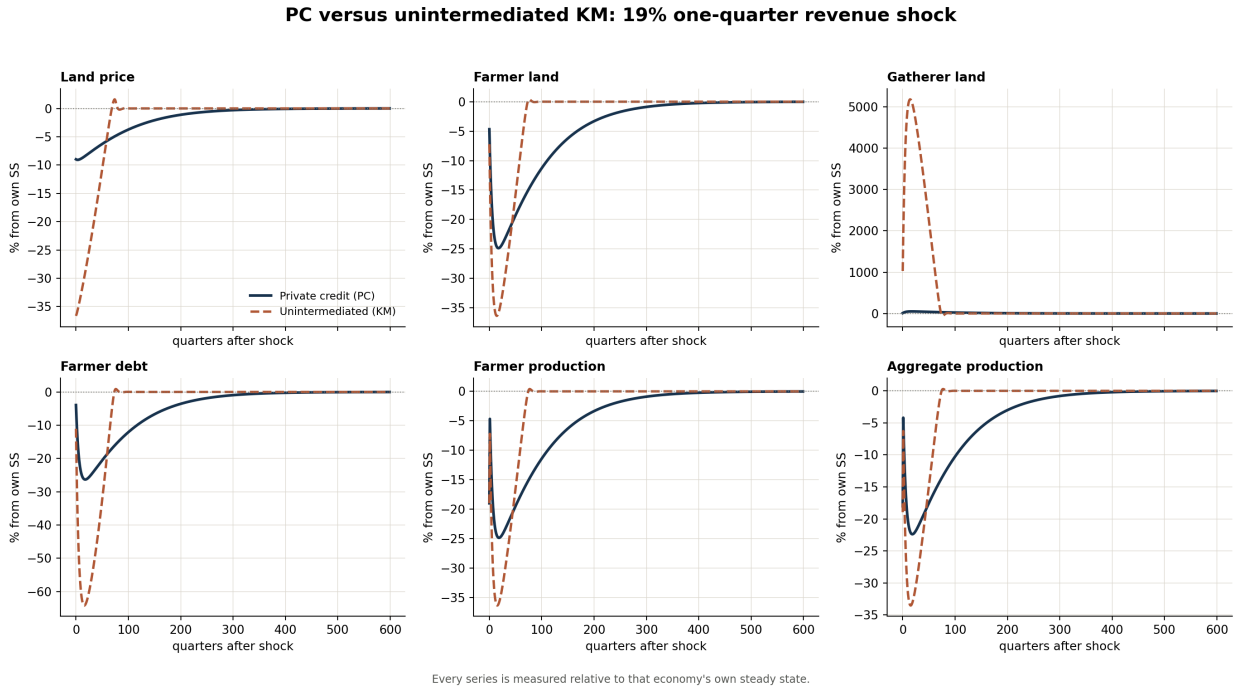


Figure 3: Response after gardener buffer exhaustion

Notes: Perfect-foresight responses to a 19% one-period decline in farmer harvest, which exceeds the gardeners' consumption buffer-exhaustion threshold. All series are measured relative to their own steady-state values.

against the transferable value of that land. The second is the corresponding own-resource requirement for the farmer loan book. Gardener consumption can therefore reach zero while the sector still has sufficient resources to support a smaller balance sheet.

Lemma 5 (Buffer exhaustion precedes gardener collapse)**.** *For the one-period shock in equation* (57)*,*

$$\Delta^A \leq \Delta^C. \tag{76}$$

If $x_0^S > 0$ after the shock, gardeners can free additional resources one-for-one by reducing consumption. They therefore cannot fail to meet inherited obligations before this margin is exhausted.

For shocks between Δ^A and Δ^C, gardener consumption is zero and further losses force gardeners to deleverage. Farmer lending falls, gardeners may sell land, and the decline in s_t reduces farmer debt capacity. At Δ^C, deleveraging is no longer sufficient to meet inherited

obligations.

In the numerical economy, gardener consumption is exhausted at a shock of about 16%, while gardener collapse occurs at roughly 29%. In the corresponding KM97 economy, the farming sector can no longer meet its inherited obligations beyond a shock of about 19%. In this example, gardener lending reduces the initial contraction and allows the economy to withstand a larger shock, at the cost of a slower recovery and lower steady-state production.

2.5 Two farmer sectors: cross-funding and flight to pledgeability

We now allow gardeners to lend to two farmer sectors. The sectors differ in the extent to which the value of cultivated land can be preserved only by gardeners. Because both sectors are financed from the same gardener balance sheet, losses in one sector can affect borrowing in the other.

We call the two farmer types orchardists, indexed by O, and croppers, indexed by C. Orchardists rely more heavily on the continuation value of installed trees, whereas a larger share of cropper collateral value is transferable to gatherers.

2.5.1 A common gardener balance sheet

For each sector $i \in \{O, C\}$, let $(a^i, \lambda^i, \phi^i, \pi^i)$ denote sector-specific production and investment parameters. Each sector has a fixed land supply $\bar{K}^i$ and land price q_t^i. Land-market clearing is

$$K_t^{F,i} + nk_t^{S,i} + mk_t^{G,i} = \bar{K}^i, \qquad i \in \{O, C\} \tag{77}$$

The farmer investment, debt, and recovery equations apply sector by sector. Let L_{t+1}^i denote gardener recovery value from one unit of cultivated type-i farmer land. Debt capacity is

$$\chi_t^i = \frac{q_{t+1}^i}{R} + s_t \left(L_{t+1}^i - q_{t+1}^i \right). \tag{78}$$

The shadow discount factor s_t is common across sectors because both loan books are financed from the same gardener balance sheet.

For sector i, let $B_t^{F,i,S}$ denote farmer loans held by gardeners and $V_{t+1}^{F,i,S}$ the value gatherers

can recover from those loans. The gardener budget constraint is

$$n \sum_{i\in\{O,C\}} q_t^i(k_t^{S,i} - k_{t-1}^{S,i}) + \sum_{i\in\{O,C\}} B_t^{F,i,S} + RB_{t-1}^S + x_t^S$$
$$= n \sum_{i\in\{O,C\}} H^i(k_{t-1}^{S,i}) + \sum_{i\in\{O,C\}} \left[V_t^{F,i,S} + \frac{B_{t-1}^{F,i,S} - V_t^{F,i,S}/R}{s_{t-1}} \right] + B_t^S \tag{79}$$

Its borrowing constraint is

$$RB_t^S \leq n \sum_{i\in\{O,C\}} q_{t+1}^i k_t^{S,i} + \sum_{i\in\{O,C\}} V_{t+1}^{F,i,S} \tag{80}$$

Define the sector-specific pledgeability wedge as

$$W_{t+1}^i \equiv L_{t+1}^i - q_{t+1}^i. \tag{81}$$

We assume

$$W_{t+1}^O > W_{t+1}^C \geq 0. \tag{82}$$

Orchardist credit therefore depends more heavily on gardener resources.

From (78),

$$\frac{\partial \chi_t^i}{\partial s_t} = W_{t+1}^i. \tag{83}$$

Lemma 6 (Differential exposure to gardener resources)**.** *A decline in the shadow discount factor s_t weakly reduces debt capacity in both sectors. The reduction is larger in the sector with the larger pledgeability wedge:*

$$\frac{\partial \chi_t^O}{\partial s_t} > \frac{\partial \chi_t^C}{\partial s_t}. \tag{84}$$

2.5.2 New lending and flight to pledgeability

The same feature that makes gardener lending especially valuable to orchardists in normal times makes them more exposed when gardener resources become scarce. A larger W_{t+1}^i means that a larger part of farmer borrowing is supported by value that cannot be pledged onward to gatherers. When s_t falls, that part of debt capacity contracts.

The effect also carries over to new lending. Let $\mathcal{B}_t^i$ denote the principal issued by the investing farmers in sector i. From the farmer policy rule,

$$\mathcal{B}_t^i = \pi^i \chi_t^i \frac{\left(a_t^i + q_t^i + \lambda^i \phi^i\right) K_{t-1}^{F,i} - R B_{t-1}^{F,i} - \left(\frac{1}{s_{t-1}} - R\right)\left(B_{t-1}^{F,i,S} - \frac{V_t^{F,i,S}}{R}\right)}{q_t^i + \phi^i - \chi_t^i}. \tag{85}$$

Holding current farmer resources and land prices fixed, the direct response of new originations to s_t is

$$\left.\frac{\partial \log \mathcal{B}_t^i}{\partial s_t}\right|_{\text{direct}} = \frac{q_t^i + \phi^i}{\chi_t^i \left(q_t^i + \phi^i - \chi_t^i\right)} W_{t+1}^i. \tag{86}$$

A larger pledgeability wedge reduces new lending through two margins. When gardener resources become scarce, debt capacity falls more, and the resulting increase in the farmer's required equity contribution further reduces the amount of land and debt that can be financed from a given amount of internal wealth.

Under local symmetry across the two sectors except for the pledgeability wedge, the right-hand side of equation (86) is increasing in W_{t+1}^i. A fall in s_t therefore reduces orchardist originations proportionately more than cropper originations. The share of new lending going to the more transferable cropper sector rises. We refer to this reallocation as a *flight to pledgeability*.

Proposition 5 (Flight to pledgeability)**.** *Suppose $W_{t+1}^O > W_{t+1}^C$. Once gardener resources become scarce, a decline in s_t reduces orchardist debt capacity more than cropper debt capacity. Conditional on borrower fundamentals, new lending falls proportionately more for orchardists than for croppers, shifting lending toward the sector with the smaller pledgeability wedge.*

2.5.3 Cross-sector transmission

Consider an unanticipated one-period productivity shock to croppers only:

$$a_0^C = (1 - \Delta) a^C, \qquad a_t^C = a^C \quad \text{for } t \geq 1, \qquad a_t^O = a^O \quad \text{for all } t. \tag{87}$$

Orchardist productivity is therefore unchanged.

For shocks small enough that gardener consumption remains positive, the common shadow

discount factor remains at β^S. The cropper shock affects orchardists only through equilibrium prices and the common gardener balance sheet. Once the shock exhausts gardener consumption, however,

$$\frac{ds_0}{d\Delta} < 0. \tag{88}$$

The effect of this common balance-sheet contraction on sector-i debt capacity is

$$W_1^i \frac{ds_0}{d\Delta} \tag{89}$$

The differential effect across sectors is therefore

$$\left(W_1^O - W_1^C\right) \frac{ds_0}{d\Delta} < 0. \tag{90}$$

Proposition 6 (Cross-sector transmission)**.** *Suppose $W_1^O > W_1^C$ and a shock to cropper productivity is sufficiently large to exhaust gardener consumption. The resulting decline in s_0 reduces orchardist debt capacity even though orchardist productivity is unchanged. Through this common balance-sheet channel, the high-wedge sector can contract more than the sector directly hit by the productivity shock.*

Figure 4 illustrates this transmission mechanism. Although only cropper productivity falls, orchardist land prices fall more, and the contractions in orchardist farmer land and debt are substantially larger. Land released by orchardist farmers is absorbed by gardeners and gatherers, whose orchardist land holdings rise by more than their cropper holdings.

The orchardist contraction does not reflect a deterioration in orchardist productivity. Rather, orchardist borrowing relies more heavily on the gardener-specific component of collateral value. The cropper shock exhausts gardener consumption and lowers s_0 for both sectors. Because $W^O > W^C$, the resulting contraction in debt capacity is larger for orchardists.

Thus, the reallocation of credit need not follow the location of the underlying productivity shock. In Figure 4, croppers are the only farmers whose productivity falls, yet scarce gardener resources shift financing away from orchardists because orchardist loans require more specialized-lender balance-sheet support.

The same mechanism applies to a shock that directly reduces gardener resources. More generally, any disturbance that lowers s_t directly reduces sector-i debt capacity in proportion

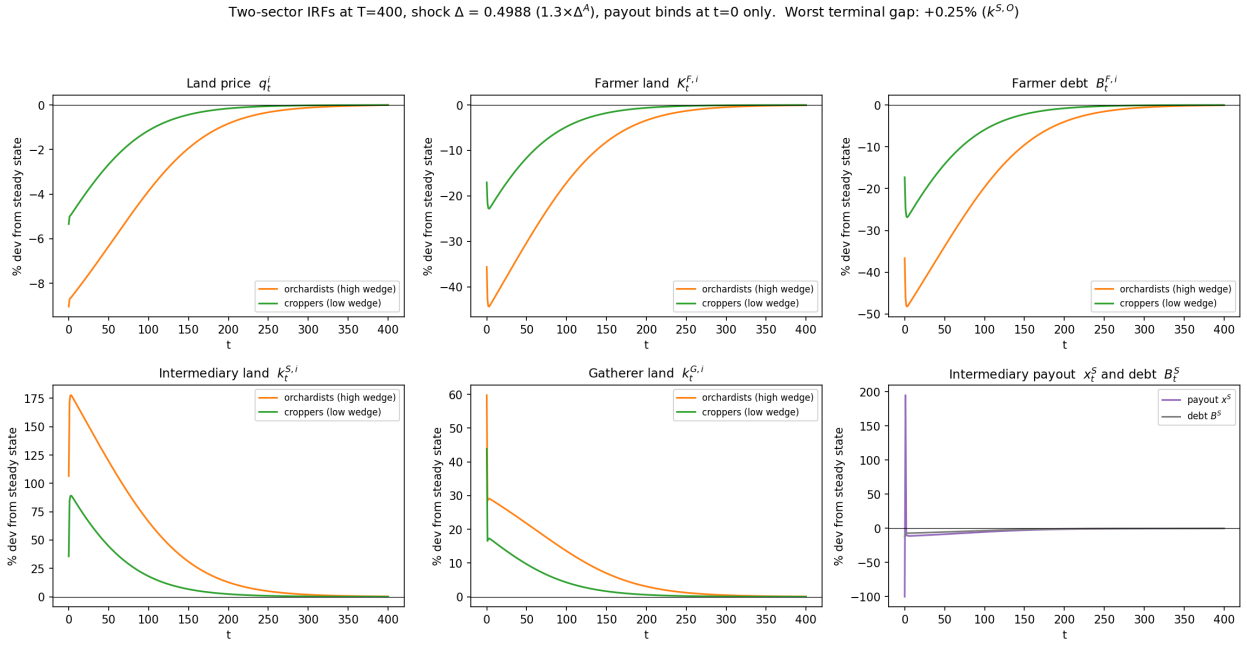


Figure 4: Cross-sector transmission of a cropper shock

Perfect-foresight responses to a one-quarter decline in cropper harvest. Orchardist productivity is unchanged. The shock is $\Delta = 0.4988$, approximately 1.3 times the buffer-exhaustion threshold in this two-sector economy, and gardener consumption is constrained at date 0 only. All series are measured relative to their own steady-state values.

to its pledgeability wedge W_{t+1}^i. This mechanism provides a balance-sheet channel through which losses in one part of a specialized lender's portfolio can affect credit supply to otherwise unrelated borrowers.

3 Facts about private capital intermediaries

We apply our model to understand the persistent growth of the private credit industry. Translating our agrarian agents into modern market participants, the farmers are the borrowing firms. The gardeners are the private credit lenders and the private equity sponsors behind them, who have the expertise to operate portfolio companies that unsophisticated lenders and investors lack. Finally, the gatherers are the rest of the savers in the economy, who do not have the expertise to run the portfolio firms independently. In this section, we

highlight four recently established facts about private credit that our model can explain. These facts correspond, in order, to four propositions of the model.

Fact 1: Direct lenders specialize in firms with intangible assets and invest heavily in monitoring and industry expertise. Dell'Ariccia et al. (2021) show that the substantial growth in intangible capital since the mid-1980s explains a large fraction of the secular decline in the share of commercial lending in banks' loan portfolios. Private credit, on the other hand, is disproportionately important for private equity-backed firms in intangible-intensive industries. Jang et al. (2025) find that the expansion of direct lending has been concentrated in enterprise-value-based lending to PE-backed firms operating in intangible-capital-intensive industries. Direct lenders write collateral claims that place greater weight on the continuation value of the firm following default, particularly in intangible industries, and are substantially more specialized by industry than banks.

Direct lenders also invest heavily in monitoring. Block et al. (2024); Jang (2024) document extensive use of financial and incurrence covenants by private debt funds. These findings suggest that the distinction between private credit and more arm's-length lending is more than the form of collateral securing the loan. Direct lenders combine concentrated ownership, information acquisition, covenants, monitoring, and industry-specific expertise in financing borrowers whose value may be difficult to realize through a simple asset sale.

Proposition 1 shows that gardeners can lend more than gatherers against the same farmer collateral because of their recovery advantage. Such advantage is greatest for firms whose continuation value is difficult to transfer to outside creditors, hence rationalizing private credit's specialization.

Fact 2: Private credit borrowers tend to be more leveraged than borrowers in the syndicated loan market. Chernenko et al. (2022) show that direct lenders lend to less profitable and more leveraged firms than the traditional broadly syndicated loans. Similarly, Banegas et al. (2026) document: 5x Debt/EBITDA-ratio for private credit borrowers versus 3.2x for broadly syndicated loan borrowers, and lower interest coverage ratio for private credit borrowers.

As shown in lemma 3 and Proposition 2, the model directly generates higher borrower leverage because specialized lenders can lend against both transferable collateral and additional lender-specific continuation value, allowing borrowers to support more debt against the same underlying project.

Fact 3: Private credit financing is less cyclical than traditional leveraged lending. Private credit, or "direct lending", becomes relatively more important during adverse credit-market conditions. Hinzen et al. (2026) document direct lending market is less cyclical than leveraged loan market. As a result, the borrowers switch to direct lending from syndicated lending when credit conditions tighten or when they experience distress.

Proposition 3 provides a mechanism for this pattern: specialized lenders can lend against going-concern value when transferable collateral values fall, cushioning the decline in borrowers' debt capacity following moderate adverse shocks.

Fact 4: Private lenders and sponsors maintain borrower relationships through distress. Direct loans are typically privately negotiated and held by the originating lender rather than distributed through an active secondary market. Because many private credit instruments lack a liquid secondary market, lenders commonly hold loans until maturity or refinancing. This ownership structure and lack of observable market prices can facilitate bilateral renegotiation when borrowers encounter financial difficulties. Recent literature has shown that this flexibility may help but sometimes make recognizing distress more gradual (Jang, 2024; Rintamäki and Steffen, 2025)[6]

The evidence on private equity-backed borrowers shows a similar preference for maintaining the firm as a going concern. Hotchkiss et al. (2021, 2023); Haque et al. (2022) find that distressed PE-backed firms restructure more quickly, are more likely to avoid bankruptcy court, and are less likely to liquidate than otherwise comparable highly leveraged firms. Sponsors are also more likely to retain control following restructuring and frequently inject additional capital as firms approach distress.

Specialized lenders can preserve going-concern value that would be lost in liquidation. Proposition 4 shows how lending against this value depends on their own resources.

Fact 5: Banks finance only part of the value of private credit loan portfolios. Bank credit agreements with business development companies (BDCs) specify an *advance rate*: the fraction of each eligible loan's assessed value available as bank financing. As an illustration, Table 1 reports Ares Capital's April 2023 advance rates, which are generally lower for junior, non-performing and unquoted claims. It is typical of the credit agreements

[6]A related development in private equity is the growing use of continuation vehicles, which allow sponsors to extend their ownership of portfolio companies beyond the typical life of the original fund (Ivashina et al., 2026; Abuzov et al., 2025). Thus, longer holding periods might allow investors to "ride out" the bad times, but this can make the recovery times more prolonged.

of BDCs.

Table 1: Example of bank advance rates on Ares Capital's portfolio assets

Portfolio asset	Quoted	Unquoted
Performing first-lien loan	85%	75%
Performing second-lien loan	75%	65%
Performing cash-pay mezzanine	65%	55%
Non-performing first-lien loan	45%	45%
Non-performing second-lien loan	40%	30%
Non-performing mezzanine	30%	25%

Source: Ares Capital Corporation, Fourteenth Amended and Restated Senior Secured Credit Agreement, April 19, 2023, §5.13, incorporated as Exhibit 10.30 in its 2023 Form 10-K. Selected rates for a relevant asset coverage ratio above 2:1, before concentration adjustments. Rates apply to values determined under §5.12, which can differ from reported fair values.

The haircut is applied after valuation: a loan assessed at 80 with a 75% advance rate supports up to 60 of bank borrowing. Even performing first-lien loans support borrowing of only 75–85% of their assessed value; junior and non-performing claims generally receive lower rates.

The model explains how such a financing gap can arise endogenously. Consider a new loan financing one unit of cultivated land, valued at its issue price. Let us assume steady state and suppress time subscripts for simplicity. Gatherers can recover only the bare-land value q, so they advance q/R. The gardener can preserve the additional recovery $L-q$, which it discounts at β_S. It therefore advances $\beta_S(L-q)$ from its own resources. The advance rate and haircut are the shares financed by gatherers and gardeners, respectively:

$$\text{Advance rate} = \frac{\overbrace{q/R}^{\text{Gatherer funding}}}{\underbrace{q/R}_{\text{Gatherer funding}} + \underbrace{\beta_S(L-q)}_{\text{Gardener resources}}}, \qquad \text{Haircut} = \frac{\beta_S(L-q)}{q/R+\beta_S(L-q)}. \tag{91}$$

For given R and β_S, the haircut increases with $(L-q)/q$.

Unquoted loans lack an active secondary market, limiting recovery through resale. Through the lens of the model, banks would therefore lend against their own recovery value. Quoted loans offer a resale option that narrows the gap with specialist recovery, consistent with their higher advance rates.

The financing haircut is typically applied to the loan's assessed fair value, which already reflects expected credit losses, and the BDC must finance the haircut itself. This suggests that higher haircuts may also reflect that these loans are harder to pledge to arm's-length lenders, which lack the expertise to capture their continuation value. Through the lens of the model, distress reduces the share of fair value that can back safe debt[7]: the pledgeable share of fair value decreases.

Overall, the evidence in Table 1 supports our argument that private credit lenders have inalienable expertise that arm's lenght lenders lack: loans whose value depends more heavily on specialist continuation activity support less bank funding per dollar of fair value.

4 Empirical evidence: COVID-19 flight to pledgeability

The two-sector model makes a sharp theoretical prediction: once gardener resources become scarce, credit contracts most in the sector with the larger pledgeability wedge (Lemma 6 and Proposition 5). In that sector, a larger share of borrowing depends on recovery value that the gardener cannot pledge to gatherers. We test this prediction using business development companies (BDCs) portfolios around COVID-19.

4.1 Data and measurement

BDCs are specialized non-bank lenders to middle-market firms. Their quarterly filings disclose their portfolios borrower by borrower in Schedules of Investments, which PitchBook compiles into security-level holdings. Our sample runs from 2018Q1 through 2024Q1. We retain debt positions with positive reported principal, exclude equity positions, and aggregate securities to the BDC–borrower–quarter level. Before the endpoint-specific restrictions described below, the resulting network contains 72 BDCs, 2,947 borrowers financed by at

[7]To see this, consider a loan financing one unit of cultivated land, contracted at $R^F b^F = L$. Suppose default occurs with probability $1/2$ and destroys half the assets – this default is caused by true distress rather than the Hart-Moore holdup problem. The gardener receives L if the loan performs and recovers $L/2$ in default. Gatherers can recover q and $q/2$, respectively. Safe gatherer funding against this loan is therefore limited to $q/(2R)$.

$$\frac{q}{q + R\beta^S(3L/2 - q)} < \frac{q}{q + R\beta^S(L - q)},$$

where the expression on the right is the advance rate before distress. Borrowing capacity falls more than fair value.

least two BDCs at some point during the sample, and 9,947 realized BDC–borrower links. The credit measure $B_{\ell bt}$ is principal outstanding, in millions of dollars. Fair value is used only to construct the lender exposure measure below.

We measure lender exposure to the COVID shock using the industry composition of each BDC's portfolio in 2019Q4. For each PitchBook IndustryCode k, let

$$C_k = -100\frac{GO_{k,2020} - GO_{k,2019}}{GO_{k,2019}}, \tag{92}$$

denote the contraction in annual BEA gross output from 2019 to 2020, after mapping the PitchBook industry classification to BEA industries. A larger C_k denotes a more adverse industry shock. BDC ℓ's exposure is

$$E_\ell = \frac{\sum_{j\in\mathcal{P}_{\ell,-1}} FV_{\ell j,-1} C_{k(j)}}{\sum_{j\in\mathcal{P}_{\ell,-1}} FV_{\ell j,-1}}, \tag{93}$$

where $\mathcal{P}_{\ell,-1}$ is the BDC's debt portfolio in 2019Q4 and $FV_{\ell j,-1}$ is the fair value of position j. Thus, E_ℓ is a 2019Q4 portfolio-weighted exposure to industries that contracted during COVID-19, measured in percentage points of output contraction. The portfolio weights are fixed before COVID-19, so subsequent lending cannot alter the exposure measure. Positions without an industry classification remain in the denominator and contribute zero to the numerator. E_ℓ measures exposure to COVID-hit industries.

The exposure sample contains 72 BDCs with positive debt portfolios in all four quarters of 2019, of which 69 enter at least one endpoint regression. Across the 72 BDCs, E_ℓ has a mean of -0.25 percentage points, a standard deviation of 3.90, and a 10th-to-90th percentile range of -4.47 to 3.24. A ten-percentage-point exposure difference, the scale used below, is therefore equal to 2.57 standard deviations.

We measure borrower pledgeability using the midpoint property, plant, and equipment recovery rate in Chapter 11 from Kermani and Ma (2023). The source measure is reported by two-digit SIC industry. We map each PitchBook borrower to its SIC industries, average the recovery rates when a borrower has more than one matched SIC industry, and then take the unweighted mean across matched borrowers within each PitchBook IndustryGroup. The borrower–SIC mapping contains 12,478 borrowers, of which 3,681 have at least one matched SIC industry and 731 have more than one. The resulting IndustryGroup measure covers

35 groups and is standardized with equal weight across those groups. Its unstandardized mean is 34.36 percent of book value and its standard deviation is 7.17 percentage points. We denote the standardized measure by $\widetilde{R}_b$. A higher $\widetilde{R}_b$ means that more collateral value is transferable in liquidation and hence that the pledgeability wedge is smaller. Table 2 reports the main summary statistics.

Table 2: Summary statistics

Variable	Observations	Mean	S.D.	10th percentile	Median	90th percentile
BDC COVID exposure (percentage points)	72	−0.25	3.90	−4.47	−0.43	3.24
IndustryGroup liquidation recovery (percent)	35	34.36	7.17	26.31	34.51	44.11
2019Q4 relationship principal ($ millions)	4,266	12.68	25.67	0.00	4.18	30.67
2021Q4 relationship principal ($ millions)	4,266	15.64	32.81	0.00	4.71	39.00
Cumulative symmetric principal growth (percentage points)	4,266	1.74	166.72	−200.00	−0.92	200.00

Notes: BDC exposure is summarized across lenders with positive debt portfolios in all four quarters of 2019. Liquidation recovery is summarized across the 35 PitchBook IndustryGroups with a mapped Kermani–Ma recovery rate. The last three rows use the fitted 2021Q4 endpoint sample after the borrower fixed-effect restriction. A zero principal at one endpoint denotes relationship entry or exit, which leads to +200 or -200 growth, respectively.

4.2 Cumulative credit-growth design

We index quarters relative to 2020Q1: $h = -1$ is the 2019Q4 baseline, $h = 0$ is 2020Q1, $h = 3$ is 2020Q4, $h = 7$ is 2021Q4, and $h = 16$ is 2024Q1. For each endpoint h, a BDC–borrower pair enters the cross-section if the BDC's full debt portfolio is observed with positive total principal in both 2019Q4 and the endpoint quarter and if $B_{\ell b,h} + B_{\ell b,-1} > 0$. This retains relationships that enter or exit between the two dates and excludes pairs with zero principal at both dates. Borrower fixed effects further require at least two usable BDC observations for a borrower in a given endpoint cross-section.

The dependent variable is the Davis–Haltiwanger–Schuh symmetric cumulative growth rate,

$$g_{\ell bh} = 100 \frac{2\,(B_{\ell b,h} - B_{\ell b,-1})}{B_{\ell b,h} + B_{\ell b,-1}}. \tag{94}$$

It equals +200 percentage points for an entering relationship and −200 percentage points for an exiting relationship. Because $B_{\ell bt}$ is principal outstanding, the outcome combines new lending and add-ons with amortization, repayment, maturity, and relationship entry or exit. It therefore measures the net adjustment in outstanding credit rather than new originations

directly.

For each endpoint h, we estimate

$$g_{\ell bh} = \gamma_h \left(\frac{E_\ell}{10}\right) + \beta_h \left(\frac{E_\ell}{10}\right) \widetilde{R}_b + \mu_{bh} + \varepsilon_{\ell bh}, \tag{95}$$

where μ_{bh} is a borrower fixed effect in endpoint h. The fixed effect structure restricts the comparison to BDCs lending to the same borrower at the same endpoint and absorbs borrower-level changes common across those lenders. The coefficient γ_h is the relative credit response to a ten-percentage-point difference in lender exposure for a borrower with mean recovery. The coefficient β_h measures how that response changes when borrower-industry recovery is one standard deviation higher.

We estimate equation (95) by weighted least squares. The weight is the relationship's baseline-endpoint midpoint share of the BDC's debt portfolio,

$$w_{\ell bh} = \frac{B_{\ell b,h} + B_{\ell b,-1}}{A_{\ell h} + A_{\ell,-1}}, \qquad A_{\ell h} = \sum_{b'} B_{\ell b',h}, \tag{96}$$

where $A_{\ell h}$ is BDC ℓ's full observed debt-portfolio principal at endpoint h. Standard errors are two-way clustered by BDC and borrower. Exposure varies only across BDCs, of which between 55 and 65 enter a given endpoint regression, and its distribution has a long right tail, so a handful of lenders carry much of the identifying variation. Inference at the BDC level therefore uses restricted, null-imposed, two-sided wild-cluster bootstrap tests with BDC bootstrap weights and the same BDC–borrower two-way studentization (Cameron et al., 2008; MacKinnon and Webb, 2017), and we report these bootstrap p-values throughout.

4.3 Results

Table 3 reports estimates at three endpoints. At the end of 2021, the lower-order exposure coefficient is $\widehat{\gamma}_7 = -40.59$ percentage points ($p = 0.005$): a more exposed BDC reduced credit relative to a less exposed BDC lending to the same borrower with mean industry recovery. The interaction coefficient is $\widehat{\beta}_7 = 25.86$ percentage points ($p = 0.007$). Thus, for two BDCs whose COVID exposures differ by ten percentage points, a one-standard-deviation increase in borrower-industry recovery makes the relative cumulative credit response 25.86 percentage

points more favorable. This is a relative reallocation result. At one standard deviation above mean recovery, the implied exposure effect remains negative at -14.73 percentage points, so the estimate means that exposed BDCs cut credit less to higher-recovery borrowers, in line with our numerical simulations and flight-to pledgeability.

The interaction is insignificant at the end of 2020 and peaks over 2021 and 2022. Lender balance sheets then recover, and the interaction fades by 2024, as the model predicts. Figure 5 shows both coefficients at every endpoint. There is no pre-trend: the seven pre-COVID coefficients are individually small and jointly insignificant ($p = 0.723$).

Table 3: Within-borrower cumulative credit growth from 2019Q4

	2020Q4 $h = 3$	2021Q4 $h = 7$	2024Q1 $h = 16$
Exposure × recovery, β_h	7.05	25.86***	10.42
	(8.35)	(9.33)	(11.58)
BDC-WCB p-value	[0.418]	[0.007]	[0.361]
Exposure, γ_h	−30.06***	−40.59***	−10.43
	(11.64)	(13.01)	(23.21)
BDC-WCB p-value	[0.002]	[0.005]	[0.802]
Observations	4,022	4,266	4,556
BDCs	65	58	55
Borrowers	1,387	1,501	1,566
IndustryGroups	33	32	32
Borrower fixed effects	Yes	Yes	Yes
Midpoint portfolio-share weights	Yes	Yes	Yes

Notes: Each column estimates equation (95) in a separate endpoint cross-section. The dependent variable is cumulative symmetric growth in BDC–borrower principal outstanding from 2019Q4 to the stated endpoint, in percentage points. The variable h denotes quarters relative to 2020Q1, so $h = -1$ is the 2019Q4 baseline. Exposure is divided by ten and recovery is standardized across IndustryGroups. Parentheses report standard errors two-way clustered by BDC and borrower. Brackets report restricted, null-imposed, two-sided wild-cluster-bootstrap p-values using 9,999 Rademacher replications, BDC bootstrap weights, and BDC–borrower two-way studentization. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

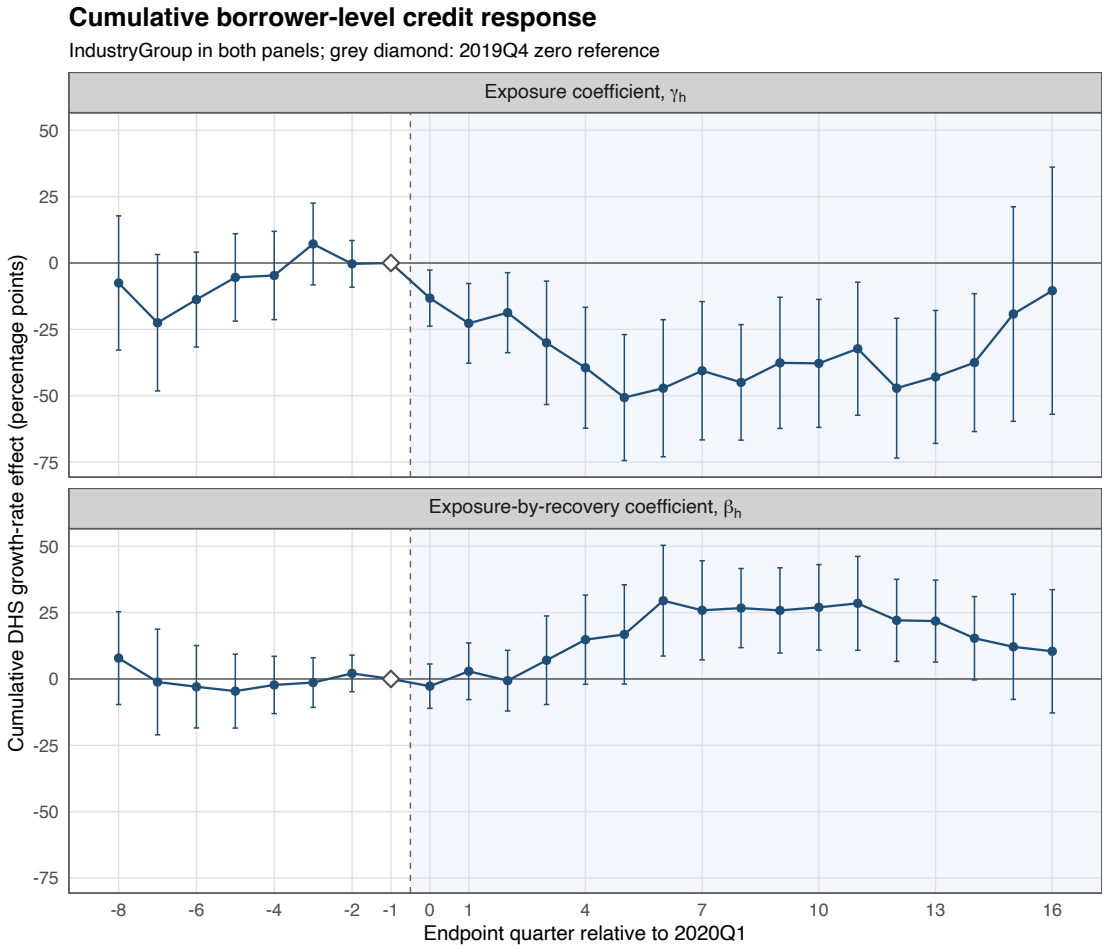


Figure 5: Cumulative within-borrower coefficients using IndustryGroup recovery. The top panel plots the exposure coefficient $\widehat{\gamma}_h$ and the bottom panel plots the exposure-by-recovery coefficient $\widehat{\beta}_h$ from equation (95). The dependent variable is symmetric growth in principal outstanding between 2019Q4 and endpoint h, where h denotes quarters relative to 2020Q1. The diamond at $h = -1$ is the mechanically zero 2019Q4 reference. Both panels show one specification, use midpoint portfolio-share weights, and report 95 percent confidence intervals based on standard errors two-way clustered by BDC and borrower.

4.4 Robustness

The main identification concern is that lender exposure may partly reflect the BDC's holdings in the focal borrower's own industry. To remove this mechanical overlap, we construct a leave-own-industry-out exposure for each BDC–borrower observation. It excludes every 2019Q4 position in the borrower's IndustryGroup from both the numerator and denominator of equation (93). The 2021Q4 interaction is then 26.09 percentage points ($p = 0.017$), compared with 25.86 in the baseline specification. The estimate is also similar when principal rather than fair value weights the 2019Q4 portfolio exposure. Appendix Figure 8 shows why the correction is small: the industries that generate exposure are high-recovery industries, so own-industry deleveraging would push the interaction toward zero.

Appendix B reports the complete robustness results and alternative recovery measures. A partial adjustment for principal reaching contractual maturity preserves a positive interaction of 20.37 percentage points, but with weaker precision ($p = 0.074$). The estimate remains positive when recovery is aggregated at the finer PitchBook IndustryCode level and when it is assigned directly from matched SIC industries. It is smaller and imprecise under equal regression weights and among relationships with positive principal at both endpoints. The reallocation therefore operates largely through relationship entry and exit rather than through the size of continuing loans, as shown in appendix B. Appendix B.6 complements this evidence at the portfolio level: more exposed BDCs shift the industry composition of their entire debt portfolios toward higher-recovery industries.

5 Conclusion

This paper develops a model of credit cycles in which specialized lenders preserve borrower going-concern value that traditional collateral-based lenders cannot fully recover. This advantage expands debt capacity and cushions credit contractions following adverse shocks, but makes lending dependent on intermediary balance-sheet resources. When these resources become scarce, credit shifts toward borrowers with more transferable collateral, generating a *flight-to-pledgeability*. We provide empirical evidence consistent with this prediction using BDC lending during the COVID-19 crisis. Our findings highlight how lender specialization can soften yet prolong credit crunches, while introducing a distinct channel of financial

amplification and credit reallocation.

Our findings have implications for the transmission of shocks in economies where specialized intermediaries play an increasingly important role. In particular, the recent expansion of private credit into asset-based lending, including financing backed by tangible assets such as data centers, creates new channels for the propagation of financial distress. Our model shows that sufficiently large losses in these relatively pledgeable investments can impair intermediary balance sheets and reduce credit supply to borrowers with less transferable collateral, even when their fundamentals remain unaffected. Thus, the diversification of specialized lenders into more tangible assets may expose their traditional borrowers to shocks originating in otherwise unrelated sectors.

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A Additional Model Derivations

This appendix provides additional derivations for the model in Section 2. We first derive the fully transferable benchmark and the pledgeable value of the gardener's farmer loan book. We then study the introduction of gardener lending into an economy initially at the steady state without gardener lending.

A.1 Fully transferable productive assets

Consider first an economy in which land is the only productive asset and is fully transferable. Because land is homogeneous and can be transferred without destroying any productive capital, the same unit of land can move freely between farmers, gardeners, and gatherers. Differences in productivity across the three groups therefore affect the competitive land price q_t rather than creating distinct physical collateral values.

Consider one unit of farmer land seized by a gardener at date $t+1$. The gardener has two options. It can sell the land immediately for q_{t+1}, or it can retain the land and operate it.

Let $\mu_t \geq 0$ denote the multiplier on the gardener funding constraint (32). Dividing the gardener borrowing condition by the marginal value of resources gives $\mu_t/\Lambda_t = 1/R - s_t$. For an interior gardener landholding, the first-order condition for land is

$$q_t = s_t \left[H'(k_t^S) + q_{t+1}\right] + (1/R - s_t)q_{t+1} \tag{97}$$

The left-hand side is the cost of one additional unit of land at date t. The first term on the right-hand side is its value to gardeners through production and resale at date $t+1$. The second term reflects the additional gatherer borrowing that the land can support.

Now consider the value, at date $t+1$, of retaining one unit of land obtained through foreclosure rather than selling it. Denote this value by $\widetilde{L}_{t+1}$. From the same gardener problem,

$$\widetilde{L}_{t+1} = s_{t+1} \left[H'(k_{t+1}^S) + q_{t+2}\right] + (1/R - s_{t+1})q_{t+2} \tag{98}$$

Keep in mind that when gardener consumption is positive at both t and $t+1$, its resources aren't scarce and $s_t = \beta^S$. Such is the case in steady state.

Equation (97), evaluated one period ahead, implies

$$\widetilde{L}_{t+1} = q_{t+1} \tag{99}$$

whenever gardeners hold land in equilibrium. The recovery value of farmer collateral to a gardener is therefore

$$L_{t+1} = \max\left\{q_{t+1}, \widetilde{L}_{t+1}\right\} = q_{t+1}. \tag{100}$$

This equality is the central result of the benchmark. A gardener may be a more productive user of land than a gatherer, but that does not create a creditor-specific recovery value. Because land is fully transferable, its productive value is already reflected in its competitive market price. A gatherer that receives the land can sell it back into that market for the same q_{t+1}.

The farmer collateral constraint consequently becomes

$$R_t^F B_t^F \leq q_{t+1} K_t^F. \tag{101}$$

Gatherers can lend directly against exactly this collateral at the safe rate R. Competition therefore implies

$$R_t^F = R. \tag{102}$$

Substituting equations (100) and (102) into the definition of debt capacity gives

$$\chi_t = \frac{q_{t+1}}{R}. \tag{103}$$

Thus, despite the presence of gardeners, the benchmark steady state coincides with the KM97 steady state in farmer production and the equilibrium land price. Because land is fully transferable, foreclosure does not create any creditor-specific recovery value: both gardeners and gatherers can ultimately realize the same market value q_t from the collateral.

A.2 Pledgeable value of the farmer loan book

Let $K_t^{F,j}$ and $B_t^{F,j}$ denote the aggregate land and net debt of farmers whose last investment occurred j periods ago. All farmers investing at date t borrow χ_t per unit of land. Between investment opportunities, harvest, land sales, and repayments are proportional to their land holdings, so farmers within each vintage continue to have the same debt-to-land ratio. Aggregate farmer land and net debt are

$$K_t^F = \sum_{j \geq 0} K_t^{F,j}, \qquad B_t^F = \sum_{j \geq 0} B_t^{F,j}. \tag{104}$$

Gardeners finance vintages whose borrowing exceeds bare-land debt capacity. Their loan book and the collateral value they can pledge to gatherers are therefore

$$B_t^{F,S} = \sum_{\substack{j\geq 0: \\ B_t^{F,j} > q_{t+1} K_t^{F,j}/R}} B_t^{F,j}, \tag{105}$$

$$V_{t+1}^{F,S} = q_{t+1} \sum_{\substack{j\geq 0: \\ B_t^{F,j} > q_{t+1} K_t^{F,j}/R}} K_t^{F,j}. \tag{106}$$

Farmers in the remaining vintages borrow from or save with gatherers. The part of the gardener loan book that cannot be financed against bare land requires gardener resources:

$$B_t^{F,S} - \frac{V_{t+1}^{F,S}}{R} = \sum_{j\geq 0} \max\left\{ B_t^{F,j} - \frac{q_{t+1} K_t^{F,j}}{R}, 0 \right\}. \tag{107}$$

Summing repayments on gardener loans and direct gatherer loans, net of returns on farmer savings, gives

$$\begin{aligned} \mathbb{E}[R_t^F b_t^F] &= V_{t+1}^{F,S} + \frac{B_t^{F,S} - V_{t+1}^{F,S}/R}{s_t} + R(B_t^F - B_t^{F,S}) \\ &= RB_t^F + \left(\frac{1}{s_t} - R\right)\left(B_t^{F,S} - \frac{V_{t+1}^{F,S}}{R}\right). \end{aligned} \tag{108}$$

The first two terms on the first line are repayments to gardeners; the last is repayment to gatherers net of returns on farmer savings. The interest premium applies only to the principal financed with gardener resources.

Since investment opportunities are iid, the newly investing vintage holds

$$K_t^{F,0} = \pi \frac{(a_t + q_t + \lambda\phi)K_{t-1}^F - RB_{t-1}^F - (1/s_{t-1} - R)\left(B_{t-1}^{F,S} - V_t^{F,S}/R\right)}{q_t + \phi - \chi_t}, \tag{109}$$

$$B_t^{F,0} = \chi_t K_t^{F,0}. \tag{110}$$

A share $1-\pi$ of each previous vintage receives no new investment opportunity. These farmers retain the fraction λ of land with surviving trees and use harvest and land-sale proceeds to

repay debt. Along an anticipated price path, their land and net debt satisfy, for $j \geq 1$,

$$K_t^{F,j} = (1-\pi)\lambda K_{t-1}^{F,j-1}, \tag{111}$$

$$B_t^{F,j} = (1-\pi)\left[RB_{t-1}^{F,j-1} - \{a_t + (1-\lambda)q_t\}K_{t-1}^{F,j-1}\right] + (1-\pi)\left(\frac{1}{s_{t-1}} - R\right)\max\left\{B_{t-1}^{F,j-1} - \frac{q_t K_{t-1}^{F,j-1}}{R}, 0\right\}. \tag{112}$$

At the unexpected date-0 shock, inherited repayments were agreed using the steady-state price q and discount factor β^S. The debt equation is therefore

$$B_0^{F,j} = (1-\pi)\left[RB_{-1}^{F,j-1} - \{a_0 + (1-\lambda)q_0\}K_{-1}^{F,j-1}\right] + (1-\pi)\left(\frac{1}{\beta^S} - R\right)\max\left\{B_{-1}^{F,j-1} - \frac{qK_{-1}^{F,j-1}}{R}, 0\right\}.$$

The land equation is unchanged. Current harvest and land sales reflect the shock; inherited repayments remain fixed.

For the newly investing vintage,

$$R_t^F B_t^{F,0} = L_{t+1}K_t^{F,0}. \tag{113}$$

When $L_{t+1} > q_{t+1}$, financing this vintage requires $s_t(L_{t+1} - q_{t+1})K_t^{F,0}$ of gardener resources.

Gardener net worth is

$$N_t^S \equiv nq_t k_t^S + B_t^{F,S} - B_t^S. \tag{114}$$

Gatherers lend to gardeners and fully covered farmers, and accept farmer savings. Their budget constraint is

$$mq_t(k_t^G - k_{t-1}^G) + B_t^S + B_t^F - B_t^{F,S} + x_t^G = mG(k_{t-1}^G) + R\left(B_{t-1}^S + B_{t-1}^F - B_{t-1}^{F,S}\right). \tag{115}$$

In the steady state considered in the main text, gardener consumption is positive, so

$$s = \beta^S. \tag{116}$$

Steady-state debt dynamics and lender allocation. In the operational foreclosure regime, $W = L - q > 0$. With positive gardener consumption, $s = \beta^S$. As long as vintage j

borrows from gardeners, its budget gives

$$\frac{B^{F,j+1}}{K^{F,j+1}} = q - \frac{a}{\lambda} + \frac{B^{F,j}/K^{F,j} - q/R}{\lambda\beta^S}. \tag{117}$$

Using the recovery value in (29) and the initial debt-to-land ratio $B^{F,0}/K^{F,0} = q/R + \beta^S W$ gives

$$\frac{B^{F,j}}{K^{F,j}} = \frac{q}{R} + \beta^S \left[W + a - a(\lambda\beta^S)^{-j}\right]. \tag{118}$$

Debt per unit of land falls with vintage age. Let J be the first vintage whose loan is fully covered by bare land. The formula above remains valid at J, so the cutoff satisfies

$$(\lambda\beta^S)^J \leq \frac{a}{a+W} < (\lambda\beta^S)^{J-1}. \tag{119}$$

Vintage J borrows or saves at the safe rate, with repayment per unit of land

$$\frac{RB^{F,J}}{K^{F,J}} = q - R\beta^S \left[a(\lambda\beta^S)^{-J} - a - W\right] \leq q \tag{120}$$

All subsequent vintages also borrow or save at R. For $j \geq J$,

$$\frac{RB^{F,j}}{K^{F,j}} = \frac{R[a + (1-\lambda)q]}{R - \lambda} + \left(\frac{R}{\lambda}\right)^{j-J} \left[\frac{RB^{F,J}}{K^{F,J}} - \frac{R[a + (1-\lambda)q]}{R - \lambda}\right]. \tag{121}$$

The operational condition implies $q < R[a+(1-\lambda)q]/(R-\lambda)$. Debt per unit of land therefore continues to fall. Farmers borrow from gatherers while debt remains positive and save with gatherers thereafter, earning R.

These policies allow farmers to repay inherited debt at each date:

$$a + (1-\lambda)q + \lambda\chi - L = a(1 - \lambda\beta^S) > 0. \tag{122}$$

The first three terms are harvest revenue, sales of land whose trees have died, and borrowing capacity against surviving cultivated land. Together they exceed L, the largest inherited repayment per unit of land.

Aggregate lending and saving. In steady state, a vintage loses the share π of farmers who invest again and the share $1 - \lambda$ of land whose trees die. Its share of farmer land is therefore

$$\frac{K^{F,j}}{K^F} = \left[1 - (1-\pi)\lambda\right] \left[(1-\pi)\lambda\right]^j. \tag{123}$$

Only vintages $0, \ldots, J-1$ borrow from gardeners. Summing their collateral, loan principal, and repayments gives

$$V^{F,S} = qK^F \left\{1 - [(1-\pi)\lambda]^J\right\}, \tag{124}$$

$$B^{F,S} = \sum_{j=0}^{J-1} K^{F,j} \left\{\frac{q}{R} + \beta^S \left[W + a - a(\lambda\beta^S)^{-j}\right]\right\}, \tag{125}$$

$$RB^{F,S} + \left(\frac{1}{\beta^S} - R\right)\left(B^{F,S} - \frac{V^{F,S}}{R}\right) = \sum_{j=0}^{J-1} K^{F,j}\left[L + a - a(\lambda\beta^S)^{-j}\right]. \tag{126}$$

Gardener resources committed to lending, per unit of farmer land, are therefore

$$\frac{B^{F,S} - V^{F,S}/R}{K^F} = \beta^S\left[1 - (1-\pi)\lambda\right]\sum_{j=0}^{J-1}\left[(1-\pi)\lambda\right]^j\left[W + a - a(\lambda\beta^S)^{-j}\right]. \tag{127}$$

For a fixed cutoff J, this expression is affine in W and hence in the land price. Appendix A.3 uses this property to show that the stationary farmer equations have at most one land price.

The remaining vintages borrow from or save with gatherers. Their aggregate net repayment is finite when $(1-\pi)R < 1$: the share of farmers who have not reinvested must decline faster than their savings grow. Under this condition,

$$\begin{aligned}\frac{R(B^F - B^{F,S})}{K^F} = {} & \left[1 - (1-\pi)\lambda\right]\left[(1-\pi)\lambda\right]^J \\ & \times \left[\frac{R[a + (1-\lambda)q]}{(R-\lambda)[1-(1-\pi)\lambda]} + \frac{RB^{F,J}/K^{F,J} - R[a+(1-\lambda)q]/(R-\lambda)}{1-(1-\pi)R}\right].\end{aligned} \tag{128}$$

Dividing by R gives their net principal per unit of farmer land. Aggregate direct gatherer lending and farmer savings are, respectively,

$$\sum_{\substack{j \geq J: \\ B^{F,j} > 0}} B^{F,j}, \qquad \sum_{\substack{j \geq J: \\ B^{F,j} > 0}} B^{F,j} - \left(B^F - B^{F,S}\right). \tag{129}$$

The first sum is finite because sufficiently old vintages become savers. Both totals follow from (121) and the land shares in (123)

Note that in the benchmark with fully transferable land, $L = q$ and farmers borrow from gatherers from the outset, so $B^{F,S} = V^{F,S} = 0$. Investing farmers borrow q/R per unit of

land. Their subsequent debt follows (121), setting $J = 0$ and $B^{F,0}/K^{F,0} = q/R$.

A.2.1 Lender allocation and repayment

A farmer whose loan repayment is fully covered by the collateral value of its land can borrow directly from a gatherer or from a gardener that finances the entire loan by borrowing from gatherers. The following lemma shows that both arrangements yield the same prices and allocations as long as all debts are repaid.

Lemma 7 (Lender allocation)**.** *Assume farmers can change lenders without cost when their loans mature, and all debts are repaid in full. A farmer loan whose repayment is fully covered by the collateral value of land can be provided directly by a gatherer or by a gardener borrowing the entire principal from gatherers. Both arrangements yield the same prices, farmer investment and debt, production, consumption, and gardener net worth. Only the lenders' gross balance sheets differ.*

The two arrangements can differ after default because only gardeners can preserve the trees. We now give a sufficient condition under which farmers initially borrowing from gatherers can repay after a shock whenever gardeners can meet their own obligations.

Start from a steady state in the operational foreclosure regime, with positive gardener consumption and a binding gardener borrowing constraint. All unsubscripted variables refer to this initial steady state. If vintage J, the first able to borrow entirely from gatherers, still has positive debt, assume

$$nH(k^S) \leq \left(q - \frac{RB^{F,J}}{K^{F,J}}\right)\left(nk^S + \frac{V^{F,S}}{q}\right). \tag{130}$$

Gardeners initially owe q per unit of land backing their borrowing, while the most indebted farmer borrowing from gatherers owes $RB^{F,J}/K^{F,J}$. The condition requires this difference to be at least as large as gardeners' harvest income per unit of backing land. Their own production therefore cannot offset their greater debt burden once the shock prevents these farmers from repaying.

Consider a one-time unexpected harvest shock with $a_0 \geq 0$, no outside equity injection, and $q_0 \geq q_1/R$. Farmers can sell their land to meet inherited obligations.

Lemma 8 (Repayment of covered loans after the shock)**.** *Under these conditions, farmers whose loan repayments were fully covered by land before the shock can repay their inherited debt for every shock below the gardener collapse threshold Δ_c.*

If $B^{F,J} \leq 0$, vintages J and above have no debt to repay. Otherwise, vintage J has the largest repayment per unit of land among farmers borrowing from gatherers. Each farmer

can raise a_0+q_0 per unit of inherited land by using its harvest and selling all its land. Failure by any of these farmers to repay therefore requires

$$a_0 + q_0 < \frac{RB^{F,J}}{K^{F,J}}. \tag{131}$$

Now consider gardeners and the farmers they financed before the shock together. These farmers hold $V^{F,S}/q$ units of land. The binding initial gardener borrowing constraint gives the group's repayment due to gatherers:

$$RB^S_{-1} = q\left(nk^S + \frac{V^{F,S}}{q}\right). \tag{132}$$

Their harvests and land can provide at most

$$nH(k^S) + a_0\frac{V^{F,S}}{q} + q_0\left(nk^S + \frac{V^{F,S}}{q}\right). \tag{133}$$

Borrowing against retained land raises only q_1/R per unit, no more than selling it at q_0. Gatherers do not lend against the additional value of preserved trees. Farmer repayments to gardeners merely transfer resources within the group, while new lending to farmers outside the group uses resources.

Using $a_0 \geq 0$, (131), and (130), we obtain

$$\begin{aligned}
&nH(k^S) + a_0\frac{V^{F,S}}{q} + q_0\left(nk^S + \frac{V^{F,S}}{q}\right) \\
&\quad \leq nH(k^S) + (a_0+q_0)\left(nk^S + \frac{V^{F,S}}{q}\right) \\
&\quad < nH(k^S) + \frac{RB^{F,J}}{K^{F,J}}\left(nk^S + \frac{V^{F,S}}{q}\right) \\
&\quad \leq q\left(nk^S + \frac{V^{F,S}}{q}\right) = RB^S_{-1}.
\end{aligned} \tag{134}$$

Thus, if a farmer initially borrowing from gatherers cannot repay, gardeners cannot repay either, even after they and their borrowers sell all their land.

The conclusion also holds if gardeners initially hold these farmer loans and finance them entirely through gatherers. Each added loan brings an equal repayment obligation to gatherers. Receiving at most the farmer's promised repayment cannot make up the shortfall.

The terms in (130) follow from the stationary vintage formulas:

$$\frac{V^{F,S}}{q} = K^F \left[1 - \{(1-\pi)\lambda\}^J\right], \tag{135}$$

$$q - \frac{RB^{F,J}}{K^{F,J}} = R\beta^S \left[a(\lambda\beta^S)^{-J} - a - (L-q)\right]. \tag{136}$$

After the initial surprise, the recovery path is anticipated, so new loans from gatherers are backed by sufficient land value at maturity. Farmers needing larger loans borrow from gardeners. Lemma 7 therefore applies throughout any recovery path with full repayment.

A.3 Gardener entry

We first compare the steady states with and without gardener lending. We then show, in a benchmark where gardeners are as patient as gatherers, that gardener-financed farmers buy more land and outcompete gatherer-financed farmers, but reach their next investment opportunity with less wealth. Finally, we compute the transition between the two steady states in the calibrated economy after gardeners enter unexpectedly.

Proof of Proposition 2. Combining the stationary farmer equations (48) and (49) gives

$$\frac{B^F}{K^F} = q + \phi - \frac{1-(1-\pi)\lambda}{\pi}\left[q + \phi - \chi\right]. \tag{137}$$

Substituting (51) and $\chi = q/R + \beta^S(L-q)$, then subtracting the corresponding equation without gardener lending, gives

$$\begin{aligned}&\left\{\pi - \left(1 - \frac{1}{R}\right)\left[1-(1-\pi)\lambda\right]\right\}(q - q^0)\\ &\quad = -\left[1-(1-\pi)\lambda\right]\beta^S(L-q) - \frac{\pi}{R-1}\left(\frac{1}{\beta^S} - R\right)\frac{B^{F,S} - V^{F,S}/R}{K^F} < 0.\end{aligned} \tag{138}$$

In the economy without gardener lending, the stationary equations also imply

$$q^0 + \phi - \frac{q^0}{R} = \frac{\pi(a + \lambda\phi)}{R\left\{\pi - (1-1/R)[1-(1-\pi)\lambda]\right\}} > 0. \tag{139}$$

The coefficient of $q - q^0$ in (138) is therefore positive, so $q < q^0$.

By (54), (55), and strict concavity of H and G, the lower land price increases gardener and gatherer landholdings. Land clearing then implies lower farmer landholdings. Since $Y^F = aK^F$ and $x^F = cK^F$, farmer production and consumption are also lower. The comparison of

borrower leverage at a given land price follows from Lemma 3. □

Corollary (uniqueness of the stationary land price). With positive gardener consumption and $(1-\pi)R < 1$, at most one land price in the operational foreclosure regime satisfies the stationary farmer equations derived above.

Proof Let q^0 be the the stationary solution to the system without gardener lending, and write $W(q) = L(q) - q$. In the operational foreclosure regime,

$$W(q) = \frac{\lambda[\beta^S a - (1 - 1/R)q]}{1 - \lambda\beta^S} > 0. \tag{140}$$

Moving all terms in (138) to the left gives

$$\begin{aligned} &\left\{\pi - \left(1 - \frac{1}{R}\right)[1 - (1-\pi)\lambda]\right\}(q - q^0) \\ &\quad + [1 - (1-\pi)\lambda]\,\beta^S W(q) + \frac{\pi}{R-1}\left(\frac{1}{\beta^S} - R\right)\frac{B^{F,S}(q) - V^{F,S}(q)/R}{K^F(q)} = 0. \end{aligned} \tag{141}$$

Equation (127) gives gardener resources committed to lending per unit of farmer land, $[B^{F,S}(q) - V^{F,S}(q)/R]/K^F(q)$. This ratio depends on the land price and the vintage distribution, but not on aggregate farmer land. Over any price interval with the same first fully covered vintage J, it is affine in $W(q)$ and therefore in q:

$$\frac{d}{dq}\left(\frac{B^{F,S}(q) - V^{F,S}(q)/R}{K^F(q)}\right) = \beta^S\left\{1 - [(1-\pi)\lambda]^J\right\}W'(q), \qquad W'(q) = -\frac{\lambda(1-1/R)}{1-\lambda\beta^S}. \tag{142}$$

The derivative of the left-hand side of (141) is therefore

$$\frac{(1-\lambda)[1-(1-\pi)R] + \pi\lambda(1 - R\beta^S)[(1-\pi)\lambda]^J}{R(1-\lambda\beta^S)} > 0. \tag{143}$$

The first term in the numerator is strictly positive because $(1-\pi)R < 1$; the second is nonnegative. When J changes, the vintage switching lenders is exactly covered by bare land and requires no gardener resources. Its contribution to (127) is therefore zero at the switch. The left-hand side of (141) is continuous and strictly increasing, and affine between switches. It has at most one zero with $L > q$. □

The condition $(1-\pi)R < 1$ also ensures finite aggregate farmer savings. To compute the price, fix J, solve (141), and check the coverage inequalities in (119). At this price, landholdings and the gardener balance sheet must still satisfy the remaining equilibrium

conditions.

A.3.1 Entry at the initial steady state

Recall that Lemma 2 gives the condition under which gardeners have a recovery advantage at the initial steady state, with L^0 given by equation (44). We assume this condition throughout the entry analysis.

For the analytical results of this subsection, we impose

$$\beta^S = \beta' = \frac{1}{R}. \tag{144}$$

Gardeners and gatherers then value future tradable fruit in the same way. Any difference between gardener and gatherer finance therefore comes from the additional recovery value $L^0 - q^0$, rather than from differences in discounting. This is the most favorable case for farmers, since with $\beta^S < \beta'$ gardener credit carries a premium. Even so, gardener lending lowers farmer wealth in the steady state.

To describe the resources of an investing farmer, define

$$e_t^F \equiv (a + q_t + \lambda\phi)\, k_{t-1}^F - R_{t-1}^F b_{t-1}^F. \tag{145}$$

This is the numerator of the farmer policy in equation (20): current harvest, the value of inherited land, and the replacement cost saved on surviving trees, net of inherited debt repayment. Consider a farmer receiving an investment opportunity at the initial steady state with internal resources $e_t^F = e > 0$.

Under gatherer finance, debt capacity is q^0/R. The farmer's required equity contribution per unit of cultivated land, as defined in equation (21), is therefore

$$q^0 + \phi - \frac{q^0}{R}. \tag{146}$$

Under gardener finance, equation (41) and the equal-discount restriction (144) imply

$$\chi = \frac{L^0}{R}. \tag{147}$$

The required farmer equity contribution therefore falls to

$$q^0 + \phi - \frac{L^0}{R}. \tag{148}$$

We assume this quantity is positive. Since $L^0 > q^0$,

$$q^0 + \phi - \frac{L^0}{R} < q^0 + \phi - \frac{q^0}{R}. \tag{149}$$

The farmer consequently acquires

$$\frac{e}{q^0 + \phi - \frac{L^0}{R}} > \frac{e}{q^0 + \phi - \frac{q^0}{R}} \tag{150}$$

units of cultivated land under gardener finance rather than gatherer finance. More precisely, the increase is

$$\frac{e}{q^0 + \phi - \frac{L^0}{R}} - \frac{e}{q^0 + \phi - \frac{q^0}{R}} = \frac{\frac{e}{R}\left(L^0 - q^0\right)}{\left(q^0 + \phi - \frac{q^0}{R}\right)\left(q^0 + \phi - \frac{L^0}{R}\right)} > 0. \tag{151}$$

Gardener finance therefore increases land purchases by a farmer receiving an investment opportunity at the prevailing price. The farmer begins with the same internal resources, but can finance a larger fraction of the acquisition and replanting cost with debt.

A.3.2 Stationary share of gardener lending

We compare steady states in which a fraction $\theta \in [0, 1]$ of investing farmers borrow from gardeners and the rest from gatherers. Farmers are assigned to lenders independently of their current internal resources.

We group farmers by the lender that financed their last investment. A farmer remains in the gardener-financed group if it refinances with gatherers before investing again.

Under the equal-discount restriction (144), both gardener-financed and gatherer-financed farmer loans carry the gross rate R. Aggregating farmer budgets across both sources of finance therefore gives the same steady-state debt-service identity as equation (49). Net farmer debt per unit of land is

$$\frac{a - (1 - \lambda)\phi}{R - 1}. \tag{152}$$

At a stationary land price q, farmers financed by gatherers contribute

$$q + \phi - \frac{q}{R} \tag{153}$$

of their own resources per unit of new cultivated land. Farmers financed by gardeners contribute

$$q + \phi - \frac{L(q)}{R}, \tag{154}$$

where, in the operational foreclosure regime,

$$L(q) = \frac{(1-\lambda)q + \frac{\lambda}{R}a}{1 - \frac{\lambda}{R}}. \tag{155}$$

Stationarity of farmer land requires

$$1 - (1-\pi)\lambda = \pi \left[a + q + \lambda\phi - R\frac{a - (1-\lambda)\phi}{R-1} \right] \times \left[\frac{1-\theta}{q + \phi - \frac{q}{R}} + \frac{\theta}{q + \phi - \frac{L(q)}{R}} \right]. \tag{156}$$

The left-hand side is the fraction of farmer cultivated land that must be replenished in each period. The first bracket on the right-hand side gives the internal resources of farmers receiving an investment opportunity per unit of inherited farmer land. The second converts these resources into newly cultivated land, taking into account that gardener-financed farmers require less equity per unit of land.

When $\theta = 0$, equation (156) reduces to the stationary condition without gardener lending and gives $q = q^0$.

Suppose the initial steady state satisfies

$$\pi > \left(1 - \frac{1}{R}\right)\left[1 - (1-\pi)\lambda\right]. \tag{157}$$

This is the local regularity condition for the stationary equilibrium branch through $(q^0, 0)$. Differentiating equation (156) with respect to θ at $\theta = 0$ gives

$$\left.\frac{dq}{d\theta}\right|_{\theta=0} = -\frac{\left[1 - (1-\pi)\lambda\right]\left(q^0 + \phi - \frac{q^0}{R}\right)}{\pi - \left(1 - \frac{1}{R}\right)\left[1 - (1-\pi)\lambda\right]} \times \left[\frac{q^0 + \phi - \frac{q^0}{R}}{q^0 + \phi - \frac{L^0}{R}} - 1 \right]. \tag{158}$$

At $\theta = 0$, changes in $L(q)$ induced by changes in q do not enter this derivative: they affect only the gardener-financed term in equation (156), whose weight is zero at the point of differentiation.

Since $L^0 > q^0$,

$$\frac{q^0 + \phi - \frac{q^0}{R}}{q^0 + \phi - \frac{L^0}{R}} > 1. \tag{159}$$

Condition (157) therefore implies

$$\left.\frac{dq}{d\theta}\right|_{\theta=0} < 0. \tag{160}$$

Proposition 7 (Stationary share of gardener lending)**.** *Suppose gardeners have a strict recovery advantage at the initial steady state, farmer equity contributions remain positive, and condition* (157) *holds. Along the stationary equilibrium branch through the initial steady state,*

$$\left.\frac{dq}{d\theta}\right|_{\theta=0} < 0. \tag{161}$$

A marginal increase in the stationary share of new farmer investments financed by gardeners therefore lowers the stationary land price.

This result contrasts with the individual response in equation (150). At the initial price, access to gardener finance allows an investing farmer to purchase more land. Yet a stationary economy in which gardener finance is used more frequently has a lower land price. The reason is that the additional borrowing changes the farmer's wealth after the initial investment.

A.3.3 Leverage displacement

An investing farmer financed by gatherers borrows q/R per unit of land, while a farmer financed by gardeners borrows $L(q)/R$. As the share of gardener finance rises, the stationary land price falls (Proposition 7). The lower price reduces the borrowing capacity of both groups, but it reduces that of gatherer-financed farmers more, because they borrow only against bare land and cannot pledge going-concern value. Gardener-financed farmers therefore outcompete gatherer-financed farmers for land.

A.3.4 Leverage and subsequent farmer wealth

Consider again a farmer with internal resources e that receives an investment opportunity at the initial steady state. Hold the land price at q^0 and compare the farmer's resources at its next investment opportunity under the two financing arrangements.

Under gatherer finance, the farmer purchases

$$\frac{e}{q^0 + \phi - \frac{q^0}{R}} \tag{162}$$

units of cultivated land. Its debt repayment is q^0 per unit. Its internal resources at the next

investment opportunity are therefore

$$e_1^G = (a + \lambda\phi)\,\frac{e}{q^0 + \phi - \frac{q^0}{R}}. \tag{163}$$

Under gardener finance, the farmer purchases

$$\frac{e}{q^0 + \phi - \frac{L^0}{R}} \tag{164}$$

units of cultivated land and repays L^0 per unit. Its internal resources at the next investment opportunity are therefore

$$e_1^S = \left[a + \lambda\phi - \left(L^0 - q^0\right)\right]\frac{e}{q^0 + \phi - \frac{L^0}{R}}. \tag{165}$$

Subtracting equations (163) and (165) gives

$$e_1^S - e_1^G = \frac{e\left(L^0 - q^0\right)}{\left(q^0 + \phi - \frac{q^0}{R}\right)\left(q^0 + \phi - \frac{L^0}{R}\right)}\left[\frac{a + \lambda\phi}{R} - \left(q^0 + \phi - \frac{q^0}{R}\right)\right]. \tag{166}$$

The initial steady-state condition implies

$$\begin{aligned} q^0 + \phi - \frac{q^0}{R} - \frac{a + \lambda\phi}{R} \\ = \frac{\frac{1}{R}\left(1 - \frac{1}{R}\right)\left[1 - (1 - \pi)\lambda\right](a + \lambda\phi)}{\pi - \left(1 - \frac{1}{R}\right)\left[1 - (1 - \pi)\lambda\right]}. \end{aligned} \tag{167}$$

Under condition (157), the right-hand side is positive. Hence

$$e_1^S < e_1^G. \tag{168}$$

Proposition 8 (Gardener finance and farmer wealth)**.** *Starting from the initial steady state, a farmer financed by gardeners acquires more land at its current investment opportunity but reaches its next investment opportunity with less internal wealth than an otherwise identical farmer financed by gatherers.*

Gardener finance has opposite effects at the current and subsequent investment opportunities. The additional recovery value lowers the farmer's required equity contribution today and allows it to purchase more land. But this expansion is financed by a larger claim on the future payoff from that land. The farmer consequently carries less internal wealth into its next investment opportunity.

Increasing θ creates newly gardener-financed farmers that purchase more land at origination, but it also increases the mass of farmers whose past gardener borrowing has left them with less internal wealth. A stationary equilibrium reflects both effects.

A.3.5 A one-time cohort experiment

Starting from the initial steady state, hold the land price fixed at q^0 and compare two small groups of otherwise identical farmers investing at date 0. One group borrows from gardeners and the other from gatherers. At subsequent investment opportunities, both groups borrow from gatherers.

Let ΔK_t^F denote the difference in farmer land attributable to this one-time change in financing. On impact,

$$\Delta K_0^F = \frac{\frac{e}{R}\left(L^0 - q^0\right)}{\left(q^0 + \phi - \frac{q^0}{R}\right)\left(q^0 + \phi - \frac{L^0}{R}\right)} > 0. \tag{169}$$

The cohort therefore initially holds more land. At subsequent investment opportunities, however, equation (168) implies that the same farmers have fewer internal resources with which to acquire and replant land.

Solving the fixed-price vintage recursion gives

$$\Delta K_t^F = \Delta K_\infty^F + \left(\Delta K_0^F - \Delta K_\infty^F\right)\left[R(1-\pi)\lambda\right]^t, \tag{170}$$

provided

$$R(1-\pi)\lambda < 1. \tag{171}$$

The long-run cohort effect satisfies

$$\Delta K_\infty^F < 0. \tag{172}$$

A.3.6 Transition after gardener entry

We now compute the equilibrium transition in the calibrated economy of Section 2.4, where $\beta^S < \beta'$. Table 4 reports the parameters. The economy starts at the steady state without gardener lending. At date 0, gardeners enter unexpectedly with initial equity E_0, and agents have perfect foresight from then on. Unlike the productivity shock in Section 2.4, entry is announced before farmers commit their date-0 labor, so they can renegotiate inherited gatherer debt down to the bare-land value of their land. We solve for the path over 800 quarters, with the steady state with gardener lending as the terminal condition.

Figure 6 shows the transition. At date 0, the land price falls by about 11%, reflecting the expected path of the economy, while farmer wealth and the distribution of debt adjust

Table 4: Calibration

Parameter	Description	Value
R	Gross safe interest rate	1.01
β	Farmer discount factor	0.98
β^S	Gardener discount factor	0.9895
β'	Gatherer discount factor	$1/R$
λ	Tree survival probability	0.93
π	Probability of an investment opportunity	0.10
ϕ	Planting cost per unit of land	12
a	Tradable harvest per unit of cultivated land	1
K	Total land	1
m, n	Masses of gatherers and gardeners	0.5, 1
$G(k) = A_G k^{\alpha_G}$	Gatherer technology	$A_G = 0.0722$, $\alpha_G = 0.6$
$H(k) = A_S k^{\alpha_S}$	Gardener technology	$A_S = 0.0291$, $\alpha_S = 0.4$
E_0	Gardener equity at entry	0.1

Notes: One period is a quarter. The numerical exercises in Section 2.4 use the same parameters.

only gradually. Following this repricing, gardener lending relaxes farmers' borrowing constraints and fuels an expansion in credit, investment and the land price: over the next 20 quarters, the land price recovers about half of its initial fall and new investment returns to its initial level, while farmer debt keeps rising for about 150 quarters. Because investment opportunities arrive intermittently and gardener equity is initially scarce, the share of farmers financed by gardeners rises only gradually. As it grows, more farmers reach their next investment opportunity with less internal wealth (Proposition 8). This cumulative wealth effect eventually overcomes the expansion. From about quarter 160, new investment, farmer debt and farmer land decline, and the economy converges to the steady state with gardener lending, where farmer land and production are about a third lower.

A.4 Additional derivations for the response to shocks

For completeness, this subsection provides additional expressions underlying the credit-cycle results in Section 2.4.

For the one-period shock considered in the main text, farmer debt capacity is

$$\chi_t = \frac{q_{t+1}}{R} + s_t \left(L_{t+1} - q_{t+1} \right). \tag{173}$$

Differentiating with respect to the size of the shock gives

$$\frac{d\chi_0}{d\Delta} = \left(\frac{1}{R} - s_0\right)\frac{dq_1}{d\Delta} + s_0\frac{dL_1}{d\Delta} + (L_1 - q_1)\frac{ds_0}{d\Delta}. \tag{174}$$

In the corresponding KM economy, farmer debt capacity is simply $\chi_0^{KM} = q_1/R$. Hence

$$\frac{d\chi_0}{d\Delta} - \frac{d\chi_0^{KM}}{d\Delta} = s_0\left(\frac{dL_1}{d\Delta} - \frac{dq_1}{d\Delta}\right) + (L_1 - q_1)\frac{ds_0}{d\Delta}. \tag{175}$$

For shocks that leave gardener consumption positive at dates 0 and 1, $ds_0/d\Delta = 0$ and $s_0 = \beta^S$, so

$$\frac{d\chi_0}{d\Delta} - \frac{d\chi_0^{KM}}{d\Delta} = \beta^S\left(\frac{dL_1}{d\Delta} - \frac{dq_1}{d\Delta}\right). \tag{176}$$

Farmer productivity returns to a at date 1. In the operational foreclosure regime, recovery is

$$L_1 = (1-\lambda)q_1 + \lambda\left[\frac{q_2}{R} + s_1(a + L_2 - q_2)\right]. \tag{177}$$

If gardener consumption is positive at dates 1 and 2, then $s_1 = \beta^S$.

Once gardener consumption is exhausted and s_0 falls with the size of the shock, equation (175) contains the additional negative term

$$(L_1 - q_1)\frac{ds_0}{d\Delta} < 0. \tag{178}$$

Since $R_0^F = L_1/\chi_0$, holding L_1 and q_1 fixed,

$$\frac{\partial R_0^F}{\partial s_0} = -\frac{L_1(L_1 - q_1)}{\chi_0^2} < 0. \tag{179}$$

These expressions make explicit the two components of the comparison with the KM economy. Before buffer exhaustion, the response differs because gardener recovery value can be less sensitive to the shock than bare-land value. After buffer exhaustion, the decline in the shadow value of future gardener resources introduces the additional balance-sheet term $(L_1 - q_1)ds_0/d\Delta$.

A.5 Additional derivation for new lending across sectors

For an investing farmer in sector i, new debt principal is $\chi_t^i k_t^{F,i}$. Aggregating across the fraction π^i of farmers receiving an investment opportunity gives

$$\mathcal{B}_t^i = \pi^i \chi_t^i \frac{\left(a_t^i + q_t^i + \lambda^i \phi^i\right) K_{t-1}^{F,i} - R B_{t-1}^{F,i} - \left(\frac{1}{s_{t-1}} - R\right)\left(B_{t-1}^{F,i,S} - \frac{V_t^{F,i,S}}{R}\right)}{q_t^i + \phi^i - \chi_t^i}. \tag{180}$$

Holding current farmer resources and land prices fixed, the direct response of new originations to s_t is

$$\left.\frac{\partial \log \mathcal{B}_t^i}{\partial s_t}\right|_{\text{direct}} = \frac{q_t^i + \phi^i}{\chi_t^i \left(q_t^i + \phi^i - \chi_t^i\right)} W_{t+1}^i. \tag{181}$$

Under local symmetry across the two sectors except for the pledgeability wedge, the right-hand side of equation (181) is increasing in W_{t+1}^i. A fall in s_t therefore reduces orchardist originations proportionately more than cropper originations. The share of new lending going to the more transferable cropper sector rises. This is the flight to pledgeability described in Section 2.5.

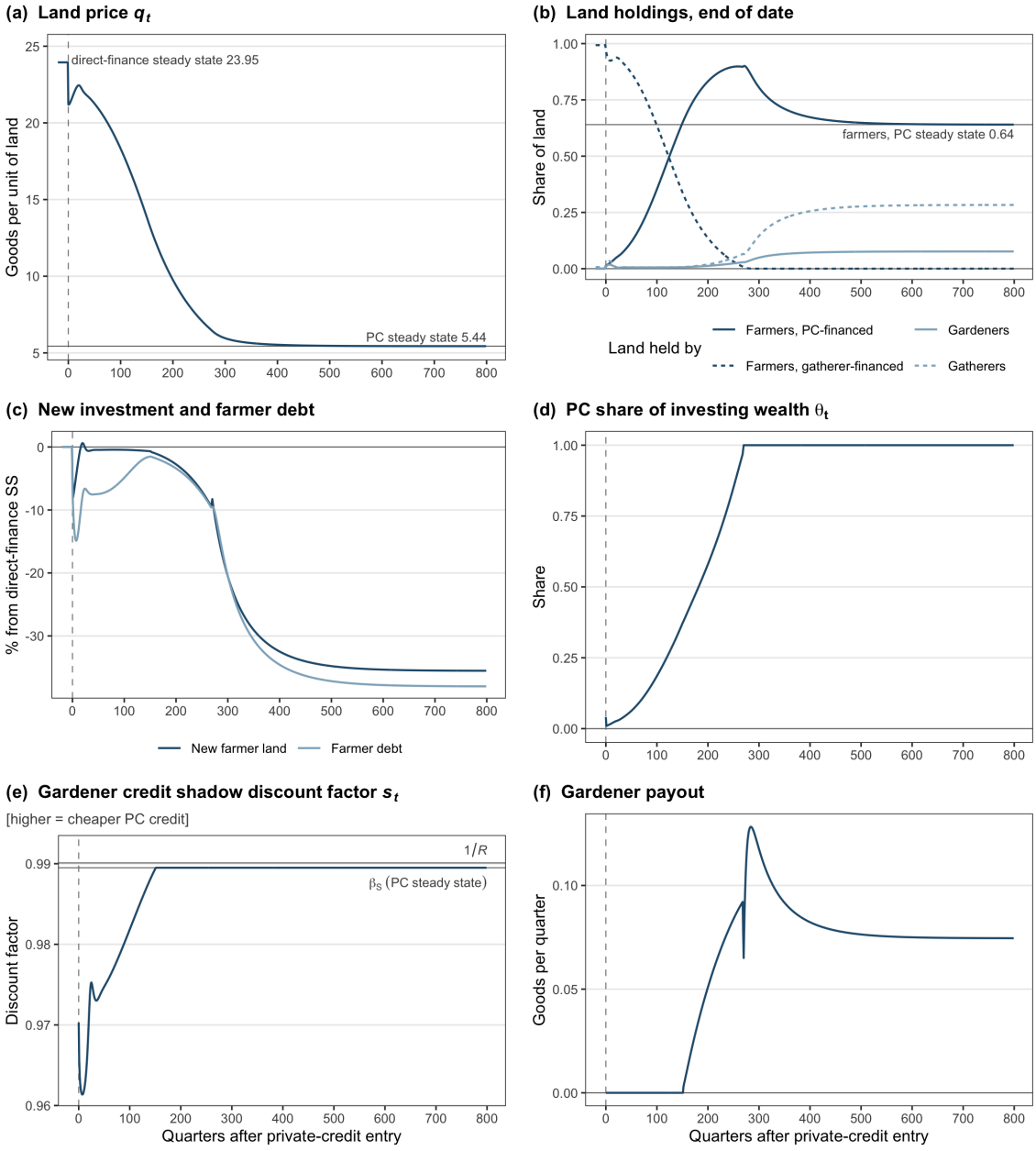


Figure 6: Transition following gardener entry

Notes: Transition from the steady state without gardener lending to the steady state with gardener lending. Gardeners enter unexpectedly at date 0 with equity $E_0 = 0.1$, and agents have perfect foresight from date 0 onwards. Panel (c) reports new farmer land and farmer debt in percent deviation from the initial steady state. Panel (d) reports the share of investing farmers' wealth financed by gardeners. From quarter 270, gardeners finance all new investment.

B Additional empirical results

B.1 Lender exposure

Figure 7 plots the distribution of E_ℓ, defined in equation (93), across the 72 BDCs with positive debt portfolios throughout 2019. Because C_k in equation (92) is negative for industries whose gross output rose in 2020, a BDC tilted toward those industries has negative exposure. The median exposure is -0.43 percentage points and the mean -0.25; the standard deviation is 3.90 and the 10th-to-90th percentile range runs from -4.47 to 3.24. Two BDCs lie far in the right tail, at roughly 10 and 19 percentage points. The ten-percentage-point exposure difference used to scale the coefficients in Table 3 equals 2.57 standard deviations of E_ℓ.

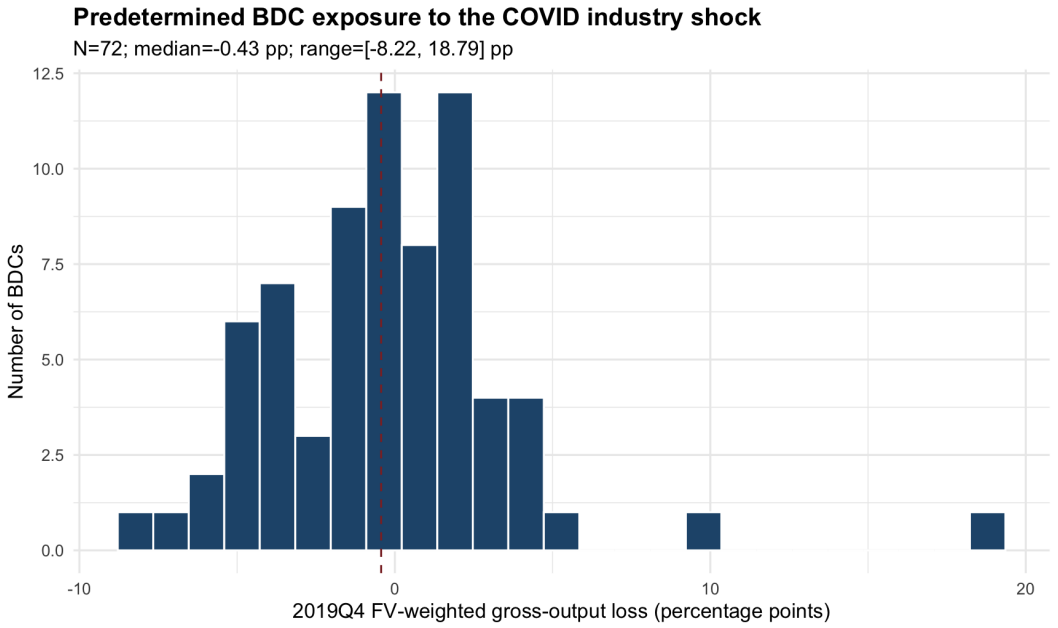


Figure 7: Distribution of predetermined BDC exposure. Histogram of E_ℓ across the 72 BDCs with positive debt portfolios in all four quarters of 2019. E_ℓ is the 2019Q4 fair-value-weighted average across BDC ℓ's debt positions of the 2019–2020 contraction in BEA gross output of the position's industry, in percentage points. The dashed line marks the median.

B.2 Industry shock and recovery

Figure 8 plots, for each of the 34 PitchBook IndustryGroups for which both measures are available, the COVID shock against the standardized recovery measure $\widetilde{R}_b$. The shock for a

group is the 2019Q4 fair-value-weighted average of C_k across the IndustryCodes it contains. The correlation is 0.51. The industries hit hardest, energy and transportation, are those with the most transferable collateral; software, healthcare technology, and pharmaceuticals combine small or negative output losses with low recovery. Restaurants, hotels and leisure is the one large-shock, low-recovery group.

The sign of this correlation rules out a mechanical reading of β_h. A BDC is exposed because of the industries it holds. If exposed BDCs simply cut the industries responsible for their losses, they would cut high-recovery borrowers by more, and the interaction would be negative. The positive β_h in Table 3 is therefore reallocation across the balance sheet, and own-industry deleveraging biases it toward zero. The leave-own-industry-out estimate in Table 6, 26.09 against 25.86 percentage points at 2021Q4, confirms that this bias is negligible.

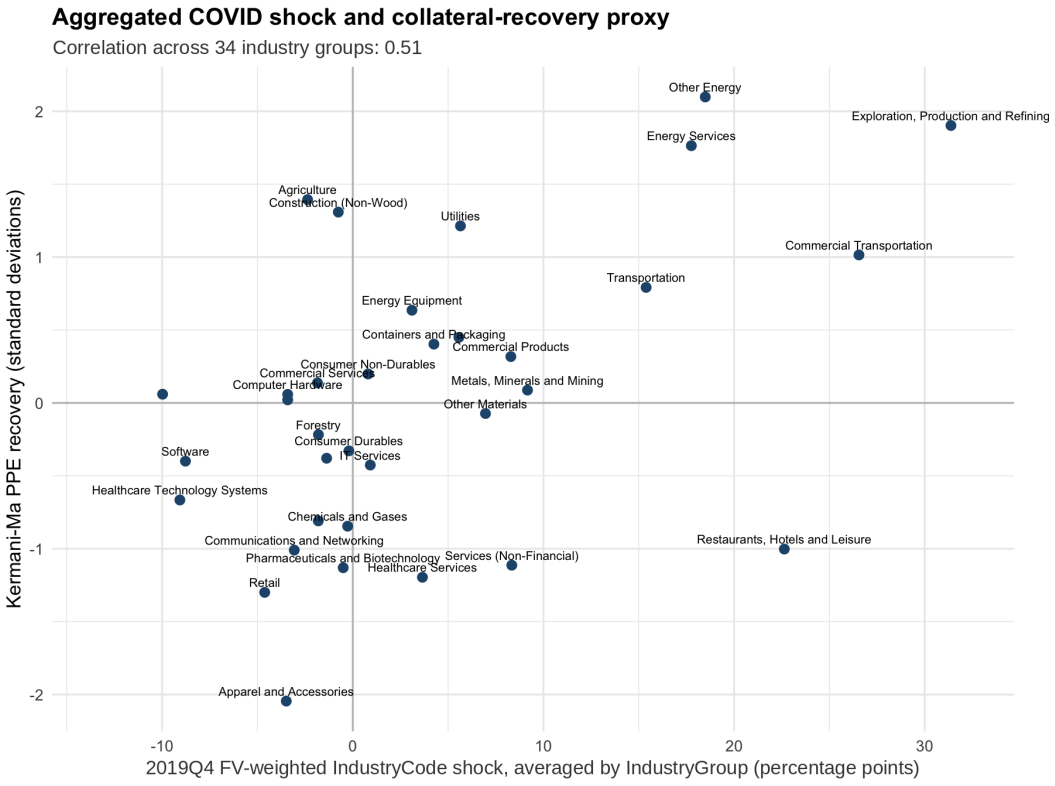


Figure 8: Industry-level COVID shock and collateral recovery. Each point is a PitchBook IndustryGroup. The horizontal axis is the 2019Q4 fair-value-weighted average across the group's IndustryCodes of the 2019–2020 contraction in BEA gross output, in percentage points. The vertical axis is the standardized Kermani and Ma (2023) property, plant, and equipment recovery rate $\widetilde{R}_b$. The correlation across the 34 groups is 0.51.

Table 5 shows the mapping between BEA codes and Pitchbook industry codes and the ranking of industry exposures to COVID-shock.

B.3 Exposure and sample robustness

Table 6 reports the 2021Q4 robustness estimates discussed in Section 4. The leave-own-industry-out measure removes the focal borrower's IndustryGroup from both the numerator and denominator of the BDC exposure measure. Principal-weighted exposure replaces the 2019Q4 fair-value weights with principal weights. Repayment-adjusted principal adds back contractual principal reaching final maturity during the outcome window. The same-industry-and-location specification compares lenders across borrowers in the same PitchBook IndustryGroup and state or province, rather than within the same borrower; it also admits single-lender borrowers, so its difference from the baseline reflects both the fixed effects and the sample. The final two rows use equal regression weights and retain only continuing relationships.

B.4 Earlier-baseline placebo

As a further check on pre-trends, we apply the same eight-quarter long-difference design to 2017Q4–2019Q4, measuring each BDC's exposure with its 2017Q4 portfolio shares in the industries subsequently hit by COVID-19. Table 7 reports the estimate. The interaction is -15.94 percentage points, with a BDC-margin wild-bootstrap p-value of 0.098. It is opposite in sign to the post-COVID estimate: over 2017Q4–2019Q4, BDCs whose portfolios were tilted toward the industries that COVID-19 would later hit grew less toward high-recovery borrowers, so a continuation of pre-COVID lending patterns would produce a negative interaction after 2019Q4, not the positive one we find. The set of multi-lender borrowers is defined over the full 2018Q1–2024Q1 sample, as in the main analysis; the placebo's exposure and outcome use no information after 2019Q4.

B.5 Alternative recovery measures

We consider two alternatives to the IndustryGroup recovery measure of the main specification. The first assigns each borrower a canonical PitchBook IndustryCode and averages directly matched borrower recovery within that code. A code uses its own mean when at least three borrowers have matched recovery and otherwise receives the mean of its parent IndustryGroup. This produces 174 IndustryCode cells, of which 137 use their own mean and 37 the parent-group mean. Recovery is then standardized with equal weight across the 174 cells.

The second alternative assigns each borrower the recovery of its matched two-digit SIC industries, averaged without weights when there are several. This measure is standardized

Table 5: BEA–PitchBook industry correspondence and COVID exposure

BEA industry	PitchBook IndustryCode	Exposure (pp)
Air transportation	Air	50.91
Support activities for mining	Other Energy Services	45.17
Accommodation	Hotels and Resorts	41.36
Petroleum and coal products	Energy Refining	33.86
Water transportation	Cruise Lines	32.55
Water transportation	Marine	32.55
Oil and gas extraction	Energy Exploration	31.33
Oil and gas extraction	Energy Production	31.33
Other transportation equipment	Aerospace and Defense	27.57
Amusements, gambling, and recreation industries	Casinos and Gaming	26.85
Amusements, gambling, and recreation industries	Leisure Facilities	26.85
Arts, entertainment, recreation, accommodation, and food services	Other Restaurants, Hotels and Leisure	25.21
Motion picture and sound recording industries	Movies, Music and Entertainment	19.09
Utilities; Oil and gas extraction	Other Energy	18.49
Food services and drinking places	Restaurants and Bars	16.73
Rail transportation	Rail	13.00
Motor vehicles, bodies and trailers, and parts	Automotive	10.32
Other services, except government	Other Consumer Products and Services	9.75
Other services, except government	Other Services (B2C Non-Financial)	9.75
Mining, except oil and gas	Aluminum Mining	9.17
Mining, except oil and gas	Coal Mining	9.17
Mining, except oil and gas	Gold Mining	9.17
Mining, except oil and gas	Iron and Steel Mining	9.17
Mining, except oil and gas	Multi-line Mining	9.17
Mining, except oil and gas	Other Metals, Minerals and Mining	9.17
Mining, except oil and gas	Precious Metals and Minerals Mining	9.17
Educational services	Education and Training Services (B2B)	8.51
Educational services	Educational and Training Services (B2C)	8.51
Nondurable goods	Other Consumer Non-Durables	7.25
Manufacturing	Other Containers and Packaging	6.96
Manufacturing	Other Materials	6.96
Fabricated metal products	Industrial Supplies and Parts	6.72
Fabricated metal products	Metal Containers and Packaging	6.72
Durable goods	Other Commercial Products	6.69
Durable goods	Other Consumer Durables	6.69
Furniture and related products	Home Furnishings	6.51
Truck transportation	Road	5.82
Utilities	Electric Utilities	5.65
Utilities	Gas Utilities	5.65
Utilities	Multi-Utilities	5.65

Table 5: BEA–PitchBook industry correspondence and COVID exposure (continued)

BEA industry	PitchBook IndustryCode	Exposure (pp)
Utilities	Other Utilities	5.65
Utilities	Water Utilities	5.65
Ambulatory health care services	Clinics/Outpatient Services	5.59
Ambulatory health care services	Laboratory Services (Healthcare)	5.59
Textile mills and textile product mills	Animal Textiles	5.57
Textile mills and textile product mills	Mineral Textiles	5.57
Textile mills and textile product mills	Other Textiles	5.57
Textile mills and textile product mills	Plant Textiles	5.57
Textile mills and textile product mills	Synthetic Textiles	5.57
Printing and related support activities	Printing Services (B2B)	4.85
Machinery	Machinery (B2B)	3.71
Machinery	Oil and Gas Equipment	3.71
Machinery	Other Equipment	3.71
Private industries	Conglomerates	3.54
Pipeline transportation	Energy Transportation	3.23
Miscellaneous professional, scientific, and technical services	Accounting, Audit and Tax Services (B2B)	3.20
Miscellaneous professional, scientific, and technical services	Accounting, Audit and Tax Services (B2C)	3.20
Miscellaneous professional, scientific, and technical services	Consulting Services (B2B)	3.20
Plastics and rubber products	Plastic Containers and Packaging	3.14
Health care and social assistance	Other Healthcare	3.03
Health care and social assistance	Other Healthcare Services	3.03
Nursing and residential care facilities	Elder and Disabled Care	2.90
Electrical equipment, appliances, and components	Alternative Energy Equipment	2.66
Electrical equipment, appliances, and components	Electrical Equipment	2.66
Electrical equipment, appliances, and components	Household Appliances	2.66
Wholesale trade	Distributors (Healthcare)	2.45
Wholesale trade	Distributors/Wholesale	2.45
Wholesale trade	Energy Marketing	2.45
Wholesale trade	Energy Traders and Brokers	2.45
Administrative and support services	BPO/Outsource Services	2.43
Administrative and support services	Human Capital Services	2.43
Administrative and support services	Office Services (B2B)	2.43
Administrative and support services	Practice Management (Healthcare)	2.43
Administrative and support services	Security Services (B2B)	2.43
Waste management and remediation services	Environmental Services (B2B)	1.97
Professional and business services	Other Business Products and Services	1.92
Professional and business services	Other Commercial Services	1.92
Legal services	Legal Services (B2B)	1.67
Legal services	Legal Services (B2C)	1.67
Computer systems design and related services	IT Consulting and Outsourcing	0.92

Table 5: BEA–PitchBook industry correspondence and COVID exposure (continued)

BEA industry	PitchBook IndustryCode	Exposure (pp)
Computer systems design and related services	Other IT Services	0.92
Computer systems design and related services	Systems and Information Management	0.92
Hospitals	Hospitals/Inpatient Services	0.81
Broadcasting and telecommunications	Broadcasting, Radio and Television	0.77
Broadcasting and telecommunications	Cable Service Providers	0.77
Broadcasting and telecommunications	Telecommunications Service Providers	0.77
Broadcasting and telecommunications	Wireless Service Providers	0.77
Food and beverage and tobacco products	Beverages	0.26
Food and beverage and tobacco products	Food Products	0.26
Chemical products	Agricultural Chemicals	-0.27
Chemical products	Biotechnology	-0.27
Chemical products	Commodity Chemicals	-0.27
Chemical products	Drug Delivery	-0.27
Chemical products	Drug Discovery	-0.27
Chemical products	Household Products	-0.27
Chemical products	Industrial Chemicals	-0.27
Chemical products	Multi-line Chemicals	-0.27
Chemical products	Other Chemicals and Gases	-0.27
Chemical products	Other Pharmaceuticals and Biotechnology	-0.27
Chemical products	Personal Products	-0.27
Chemical products	Pharmaceuticals	-0.27
Chemical products	Specialty Chemicals	-0.27
Forestry, fishing, and related activities	Aquaculture	-0.37
Forestry, fishing, and related activities	Forestry Development/Harvesting	-0.37
Paper products	Paper Containers and Packaging	-0.48
Paper products	Paper/Soft Products	-0.48
Nonmetallic mineral products	Building Products	-0.76
Nonmetallic mineral products	Raw Materials (Non-Wood)	-0.76
Management of companies and enterprises	Holding Companies	-1.13
Federal Reserve banks, credit intermediation, and related activities	Consumer Finance	-1.30
Federal Reserve banks, credit intermediation, and related activities	International Banks	-1.30
Federal Reserve banks, credit intermediation, and related activities	National Banks	-1.30
Federal Reserve banks, credit intermediation, and related activities	Other Commercial Banks	-1.30
Federal Reserve banks, credit intermediation, and related activities	Regional Banks	-1.30
Federal Reserve banks, credit intermediation, and related activities	Specialized Finance	-1.30
Federal Reserve banks, credit intermediation, and related activities	Thrifts and Mortgage Finance	-1.30
Warehousing and storage	Energy Storage	-1.57
Government	Government	-1.66
Real estate	Buildings and Property	-1.77
Real estate	Real Estate Investment Trusts (REITs)	-1.77

Table 5: BEA–PitchBook industry correspondence and COVID exposure (continued)

BEA industry	PitchBook IndustryCode	Exposure (pp)
Real estate	Real Estate Services (B2C)	-1.77
Agriculture, forestry, fishing, and hunting	Other Agriculture	-2.24
Wood products	Forestry Processing	-2.26
Wood products	Wood Containers and Packaging	-2.26
Wood products	Wood/Hard Products	-2.26
Farms	Animal Husbandry	-2.50
Farms	Cultivation	-2.50
Farms	Horticulture	-2.50
Information	Other Communications and Networking	-2.82
Information	Other Media	-2.82
Other transportation and support activities	Infrastructure	-3.40
Other transportation and support activities	Logistics	-3.40
Other transportation and support activities	Other Transportation	-3.40
Computer and electronic products	Application Specific Semiconductors	-3.41
Computer and electronic products	Computers, Parts and Peripherals	-3.41
Computer and electronic products	Connectivity Products	-3.41
Computer and electronic products	Electronic Components	-3.41
Computer and electronic products	Electronic Equipment and Instruments	-3.41
Computer and electronic products	Electronics (B2C)	-3.41
Computer and electronic products	Fiberoptic Equipment	-3.41
Computer and electronic products	General Purpose Semiconductors	-3.41
Computer and electronic products	Office Electronics	-3.41
Computer and electronic products	Other Hardware	-3.41
Computer and electronic products	Other Semiconductors	-3.41
Computer and electronic products	Production (Semiconductors)	-3.41
Computer and electronic products	Storage (IT)	-3.41
Computer and electronic products	Wireless Communications Equipment	-3.41
Apparel and leather and allied products	Accessories	-3.49
Apparel and leather and allied products	Clothing	-3.49
Apparel and leather and allied products	Footwear	-3.49
Apparel and leather and allied products	Luxury Goods	-3.49
Apparel and leather and allied products	Other Apparel	-3.49
Other retail	Catalog Retail	-3.80
Other retail	Internet Retail	-3.80
Other retail	Other Retail	-3.80
Other retail	Specialty Retail	-3.80
Construction	Construction and Engineering	-3.84
Construction	Energy Infrastructure	-3.84
Finance and insurance	Other Financial Services	-4.13
Securities, commodity contracts, and investments	Asset Management	-4.21

Table 5: BEA–PitchBook industry correspondence and COVID exposure (continued)

BEA industry	PitchBook IndustryCode	Exposure (pp)
Securities, commodity contracts, and investments	Brokerage	-4.21
Securities, commodity contracts, and investments	Investment Banks	-4.21
Securities, commodity contracts, and investments	Other Capital Markets/Institutions	-4.21
Insurance carriers and related activities	Automotive Insurance	-5.39
Insurance carriers and related activities	Commercial/Professional Insurance	-5.39
Insurance carriers and related activities	Insurance Brokers	-5.39
Insurance carriers and related activities	Life and Health Insurance	-5.39
Insurance carriers and related activities	Managed Care	-5.39
Insurance carriers and related activities	Multi-line Insurance	-5.39
Insurance carriers and related activities	Other Insurance	-5.39
Insurance carriers and related activities	Property and Casualty Insurance	-5.39
Insurance carriers and related activities	Re-Insurance	-5.39
General merchandise stores	Department Stores	-6.64
General merchandise stores	General Merchandise Stores	-6.64
Publishing industries, except internet (includes software)	Application Software	-8.78
Publishing industries, except internet (includes software)	Automation/Workflow Software	-8.78
Publishing industries, except internet (includes software)	Business/Productivity Software	-8.78
Publishing industries, except internet (includes software)	Communication Software	-8.78
Publishing industries, except internet (includes software)	Database Software	-8.78
Publishing industries, except internet (includes software)	Educational Software	-8.78
Publishing industries, except internet (includes software)	Enterprise Systems (Healthcare)	-8.78
Publishing industries, except internet (includes software)	Entertainment Software	-8.78
Publishing industries, except internet (includes software)	Financial Software	-8.78
Publishing industries, except internet (includes software)	Internet Software	-8.78
Publishing industries, except internet (includes software)	Medical Records Systems	-8.78
Publishing industries, except internet (includes software)	Multimedia and Design Software	-8.78
Publishing industries, except internet (includes software)	Network Management Software	-8.78
Publishing industries, except internet (includes software)	Operating Systems Software	-8.78
Publishing industries, except internet (includes software)	Other Software	-8.78
Publishing industries, except internet (includes software)	Publishing	-8.78
Publishing industries, except internet (includes software)	Social/Platform Software	-8.78
Publishing industries, except internet (includes software)	Software Development Applications	-8.78
Publishing industries, except internet (includes software)	Vertical Market Software	-8.78
Miscellaneous manufacturing	Business Equipment and Supplies	-9.98
Miscellaneous manufacturing	Diagnostic Equipment	-9.98
Miscellaneous manufacturing	Discovery Tools (Healthcare)	-9.98
Miscellaneous manufacturing	Medical Supplies	-9.98
Miscellaneous manufacturing	Monitoring Equipment	-9.98
Miscellaneous manufacturing	Other Devices and Supplies	-9.98
Miscellaneous manufacturing	Recreational Goods	-9.98

Table 5: BEA–PitchBook industry correspondence and COVID exposure (continued)

BEA industry	PitchBook IndustryCode	Exposure (pp)
Miscellaneous manufacturing	Surgical Devices	-9.98
Miscellaneous manufacturing	Therapeutic Devices	-9.98
Data processing, internet publishing, and other information services	Decision/Risk Analysis	-12.26
Data processing, internet publishing, and other information services	Information Services (B2C)	-12.26
Data processing, internet publishing, and other information services	Internet Service Providers	-12.26
Data processing, internet publishing, and other information services	Media and Information Services (B2B)	-12.26
Data processing, internet publishing, and other information services	Other Healthcare Technology Systems	-12.26
Data processing, internet publishing, and other information services	Other Information Technology	-12.26
Data processing, internet publishing, and other information services	Outcome Management (Healthcare)	-12.26
Data processing, internet publishing, and other information services	Social Content	-12.26
Funds, trusts, and other financial vehicles	Private Equity	-12.94
Funds, trusts, and other financial vehicles	Securities & Trusts	-12.94
Funds, trusts, and other financial vehicles	Special Purpose Acquisition Company (SPAC)	-12.94
Funds, trusts, and other financial vehicles	Structured Finance (CLO)	-12.94

Exposure is $-100 \times (GO_{2020} - GO_{2019})/GO_{2019}$, where GO denotes annual BEA gross output. Positive values indicate contractions; negative values indicate growth. Each PitchBook IndustryCode receives the exposure of its BEA category in the canonical IndustryCode-to-BEA crosswalk. Where multiple BEA categories appear, they are separated by semicolons in the first column and exposure is their arithmetic mean. All 214 crosswalk codes are included, including broad mappings. No BDC portfolio weights are applied. Rows are ranked by unrounded exposure from highest to lowest; ties are ordered alphabetically by PitchBook IndustryCode. Values are percentage points, rounded to two decimals. Source: BEA, `GrossOutput_Ind_post97.xls`, first worksheet, and the project's canonical PitchBook IndustryCode-to-BEA crosswalk.

Table 6: Cumulative credit growth to 2021Q4: robustness

	$\widehat{\beta}_7$	S.E.	BDC-WCB p	Observations
Baseline exposure	25.86	9.33	0.007	4,266
Leave-own-IndustryGroup-out exposure	26.09	11.64	0.017	4,266
Principal-weighted exposure	23.57	8.34	0.010	4,266
Repayment-adjusted principal	20.37	10.58	0.074	4,266
Same-industry-and-location comparison	10.26	10.67	0.368	6,768
Equal regression weights	15.27	9.67	0.187	4,266
Continuing relationships only	4.69	6.74	0.508	1,250

Notes: The table reports the exposure-by-recovery coefficient at $h = 7$. All regressions include the lower-order exposure term. Except in the equal-weight row, observations receive baseline-endpoint midpoint portfolio-share weights. Standard errors are two-way clustered by BDC and borrower. BDC-WCB p-values are two-sided wild-cluster-bootstrap tests with 9,999 Rademacher replications, BDC bootstrap weights, and BDC–borrower two-way studentization. Leave-own-IndustryGroup-out exposure removes the focal borrower's IndustryGroup from both the numerator and denominator of the BDC exposure measure. Principal-weighted exposure replaces fair-value weights with par-value principal weights. Repayment-adjusted principal adds contractual principal reaching final maturity during the outcome window; it does not adjust for interim amortization, voluntary prepayment, or refinancing. The same-industry-and-location row replaces borrower fixed effects with PitchBook IndustryGroup-by-state-or-province fixed effects and admits single-lender borrowers. All other rows include borrower fixed effects. The final row retains only relationships with positive principal in both 2019Q4 and 2021Q4.

Table 7: Horizon-matched pre-COVID placebo

	2017Q4–2019Q4
Exposure × recovery	−15.94*
BDC–borrower clustered S.E.	(7.95)
Conventional p-value	[0.049]
BDC-WCB p-value	[0.098]
Observations	4,498
BDCs	63
Borrowers	1,484
Borrower fixed effects	Yes
Midpoint portfolio-share weights	Yes

Notes: The outcome is symmetric principal growth from 2017Q4 to 2019Q4, matching the eight-quarter length of the main 2019Q4–2021Q4 comparison. Exposure uses 2017Q4 fair-value portfolio shares applied to the eventual 2019–2020 industry output shocks. Recovery is the standardized IndustryGroup measure. The BDC-WCB test uses BDC–borrower studentization and 9,999 BDC-level Rademacher draws. The star uses the bootstrap p-value: $^{*}p < 0.10$.

across the 3,681 borrowers with a direct match and leaves unmatched borrowers out. Table 8 reports the corresponding within-borrower estimates, and Figures 9 and 10 the complete coefficient paths. The IndustryCode path is indistinguishable from the baseline. The direct SIC measure, on a sample less than half the size, gives a larger and less precise interaction at 2021Q4, 35.79 percentage points with a bootstrap p-value of 0.060, and a larger exposure main effect.

Table 8: Within-borrower estimates using alternative recovery measures

	2020Q4 $h = 3$	2021Q4 $h = 7$	2024Q1 $h = 16$
Panel A: IndustryCode recovery			
Exposure × recovery, β_h	5.35	25.28***	6.26
	(7.60)	(9.43)	(12.75)
BDC-WCB p-value	[0.497]	[0.010]	[0.700]
Exposure, γ_h	−30.19***	−41.93***	−11.08
	(11.70)	(13.20)	(23.77)
BDC-WCB p-value	[0.002]	[0.004]	[0.793]
Observations	4,022	4,266	4,556
Borrowers	1,387	1,501	1,566
Panel B: Direct SIC recovery			
Exposure × recovery, β_h	−8.94	35.79*	21.68
	(13.81)	(16.30)	(17.48)
BDC-WCB p-value	[0.534]	[0.060]	[0.209]
Exposure, γ_h	−38.27***	−75.66***	−9.02
	(15.16)	(18.93)	(29.13)
BDC-WCB p-value	[0.002]	[< 0.001]	[0.865]
Observations	2,126	2,010	2,041
Borrowers	643	632	638
Borrower fixed effects	Yes	Yes	Yes
Midpoint portfolio-share weights	Yes	Yes	Yes

Notes: Each column is a separate endpoint cross-section. Panel A aggregates and standardizes recovery across PitchBook IndustryCodes. Panel B assigns recovery directly from matched two-digit SIC industries and standardizes it across matched borrowers. All other definitions and inference follow Table 3.

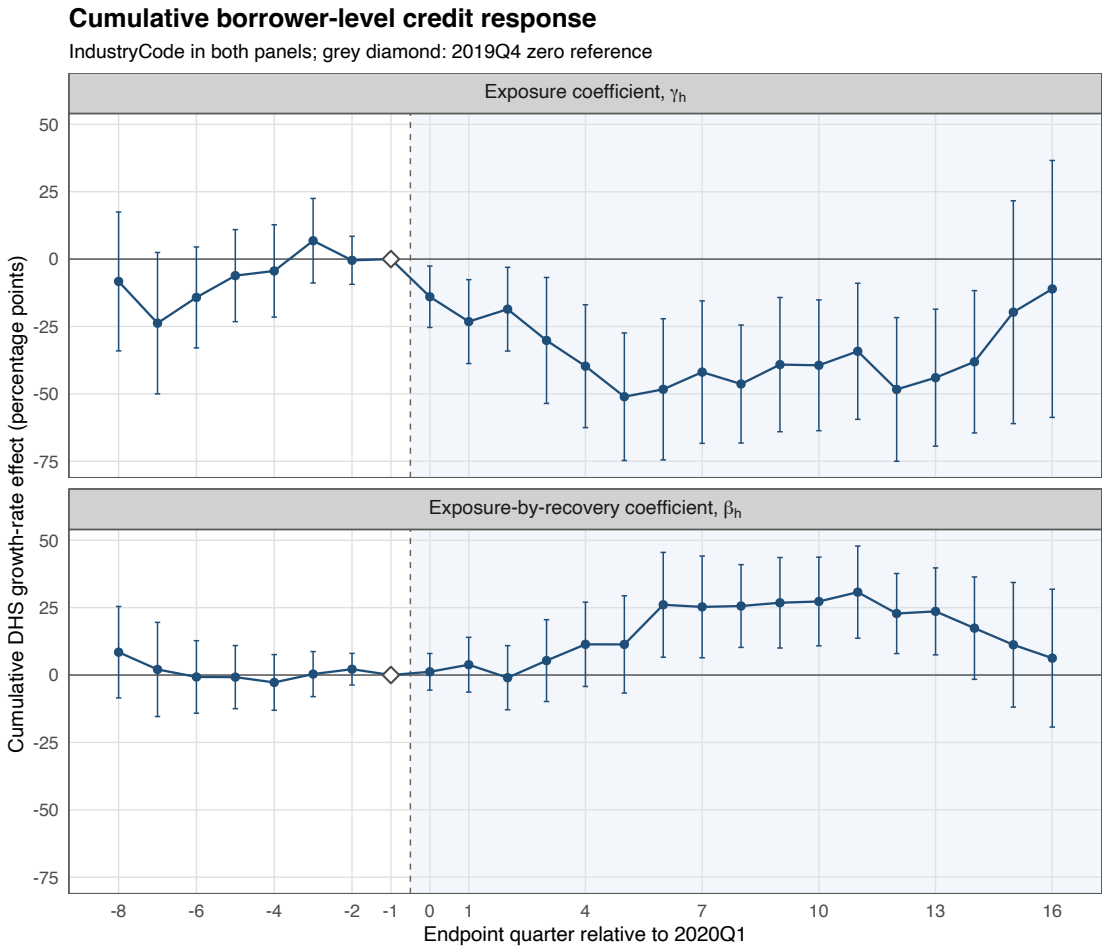


Figure 9: Cumulative within-borrower coefficients using IndustryCode recovery. The top panel plots the exposure coefficient $\widehat{\gamma}_h$ and the bottom panel plots the exposure-by-recovery coefficient $\widehat{\beta}_h$. The specification uses midpoint portfolio-share weights and 95 percent confidence intervals based on standard errors two-way clustered by BDC and borrower.

B.6 Portfolio-share reallocation

The within-borrower design of Section 4 relies on borrowers with several BDC lenders. As a complement, we ask whether more exposed BDCs shifted their whole portfolios toward higher-recovery industries. For each BDC and industry, we compute the industry's share of the BDC's debt principal in each quarter and measure its change relative to the 2019 average, either as a difference in percentage points or as a symmetric growth rate defined as in equation (94), with shares in place of principal. We regress these changes on $(E_\ell/10)\widetilde{R}_k$,

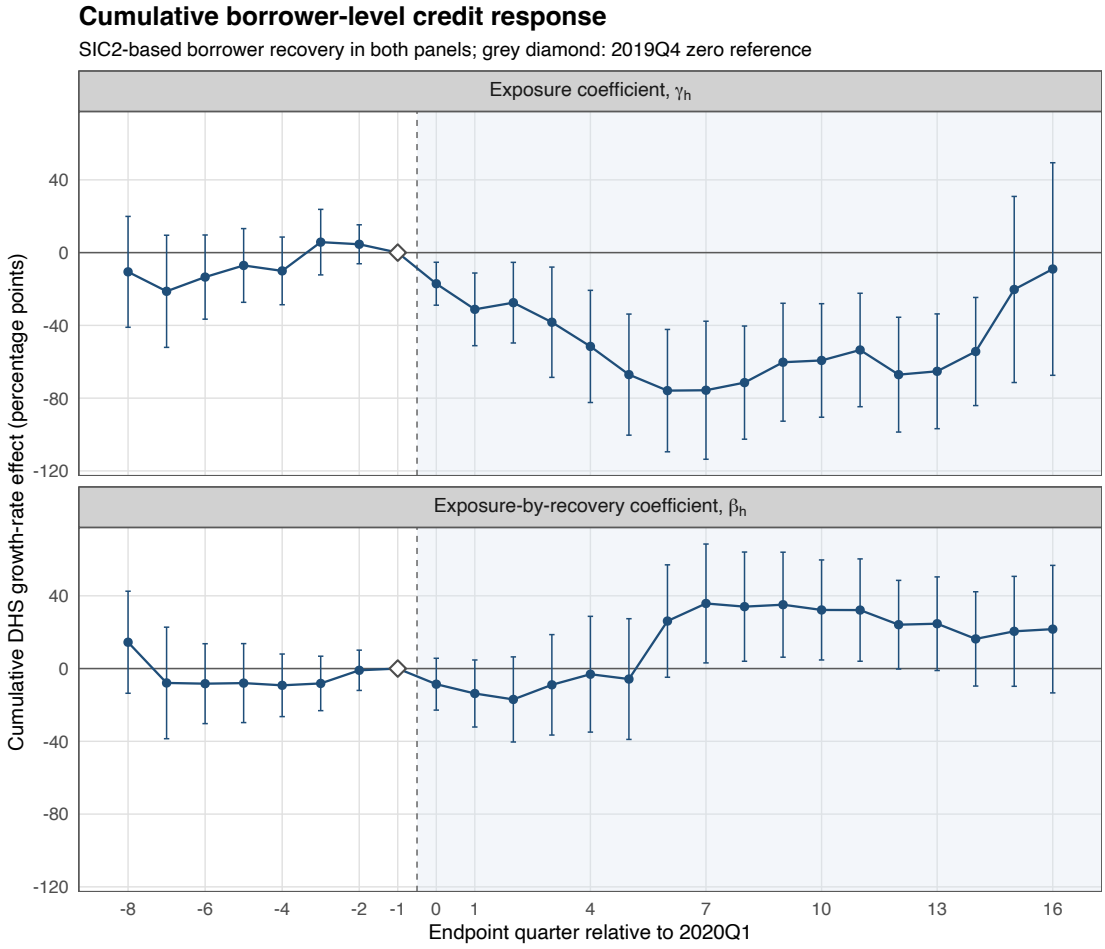


Figure 10: Cumulative within-borrower coefficients using direct SIC-based borrower recovery. The top panel plots the exposure coefficient $\widehat{\gamma}_h$ and the bottom panel plots the exposure-by-recovery coefficient $\widehat{\beta}_h$. The specification uses midpoint portfolio-share weights and 95 percent confidence intervals based on standard errors two-way clustered by BDC and borrower.

where $\widetilde{R}_k$ is the standardized recovery rate of industry k, with BDC-by-quarter and industry-by-quarter fixed effects. Because industry shares mean-revert strongly (Table 10), each regression also controls for the 2019 share, with a separate coefficient in each quarter. As in Section 4, the wild-cluster bootstrap is clustered along the dimension with the fewest effective clusters, here industries. Flight to pledgeability predicts a positive coefficient.

Table 9 pools the quarters from 2020Q2 to 2021Q3, and Figures 11 and 12 report the coefficient quarter by quarter. The pooled coefficient is positive in every specification. It is significant when recovery is measured at the IndustryGroup level, as in Section 4, and imprecise at the finer IndustryCode level.

Table 9: Portfolio-share reallocation after the COVID shock

	Change in share		Growth in share	
	IndustryCode	IndustryGroup	IndustryCode	IndustryGroup
Exposure × recovery	0.044	0.211*	12.81	23.66**
	(0.031)	(0.102)	(7.61)	(7.55)
Industry-WCB p-value	[0.248]	[0.081]	[0.125]	[0.031]
Observations	70,992	14,280	18,762	7,687
BDCs	72	72	72	72
Industries	174	35	161	33
2019 share × quarter	Yes	Yes	Yes	Yes
BDC × quarter fixed effects	Yes	Yes	Yes	Yes
Industry × quarter fixed effects	Yes	Yes	Yes	Yes
Midpoint share weights	No	No	Yes	Yes

Notes: Pooled regressions over 2020Q2–2021Q3, with the BDC sample of Section 4. The coefficient compares the response to a 10-percentage-point difference in exposure across industries one standard deviation apart in recovery. The growth columns are weighted by the average of the 2019 and current shares and exclude cells with zero shares in both periods. IndustryGroup recovery is defined as in Section 4 and IndustryCode recovery as in Appendix B.5. Standard errors in parentheses are two-way clustered by BDC and industry. Brackets report wild-cluster bootstrap p-values with 9,999 industry-level draws; stars are based on these p-values, with * and ** denoting significance at 10% and 5%.

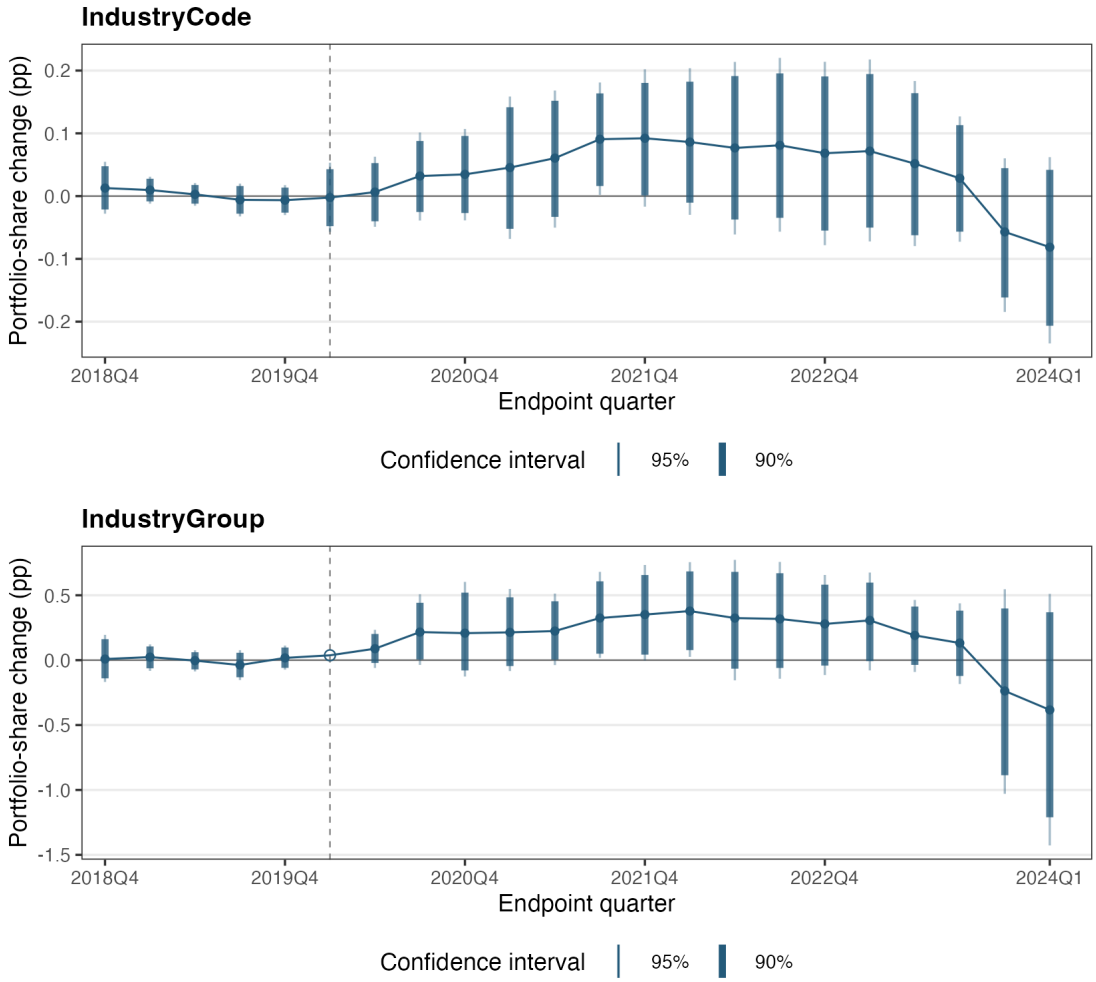


Figure 11: Change in industry portfolio shares by endpoint quarter. Each point is the exposure-by-recovery coefficient from a separate regression for that quarter, with the specification of the first two columns of Table 9. Bars show 90 and 95 percent confidence intervals obtained by inverting the industry-level bootstrap tests. The dashed line marks 2020Q1, where the IndustryGroup estimate has no interval because its two-way clustered variance estimate is negative.

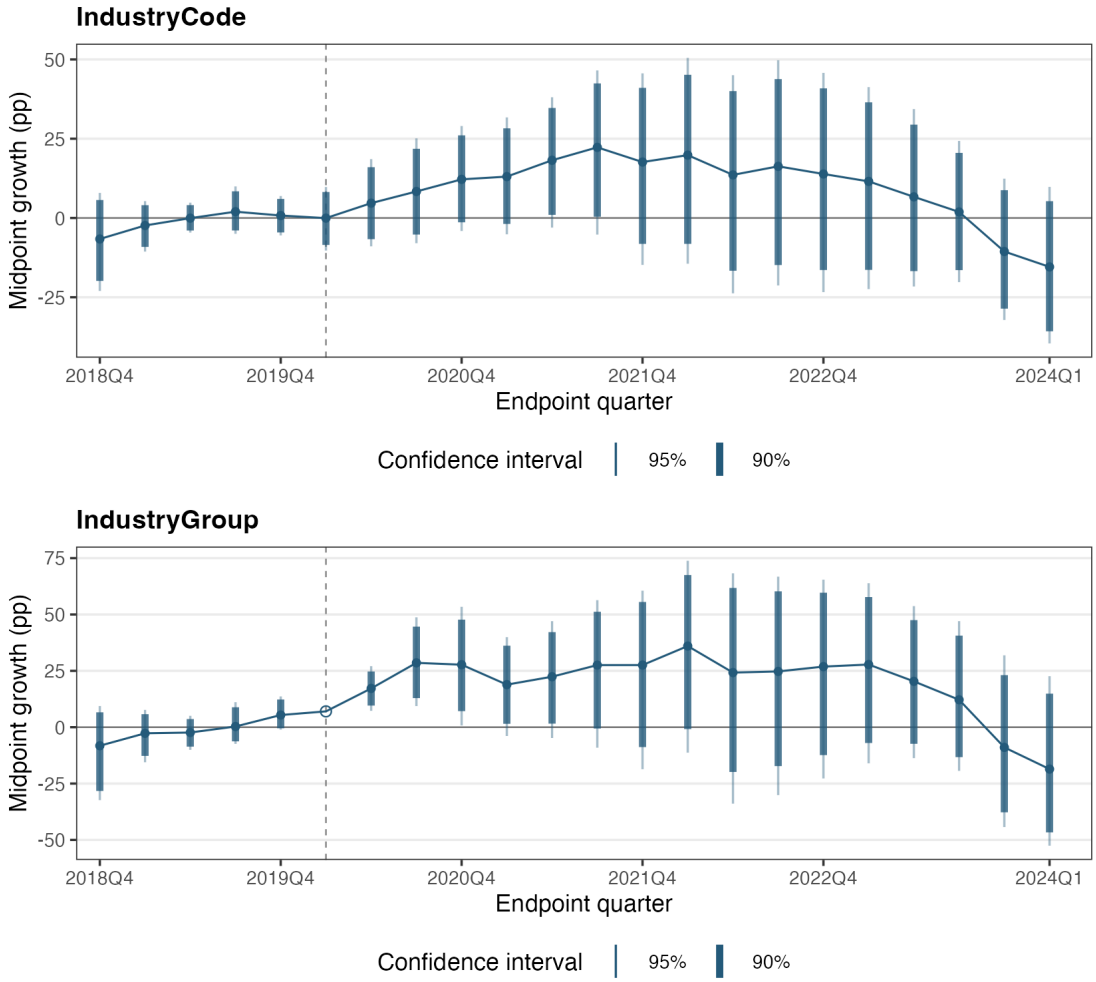


Figure 12: Growth in industry portfolio shares by endpoint quarter. Each point is the exposure-by-recovery coefficient from a separate regression for that quarter, with the specification of the last two columns of Table 9. Intervals and markers are as in Figure 11.

Table 10: Mean reversion in BDC industry portfolio shares

	IndustryGroup			IndustryCode		
Years after baseline	1	2	4	1	2	4
Baseline industry share	-0.259^{***}	-0.355^{***}	-0.480^{***}	-0.342^{***}	-0.462^{***}	-0.591^{***}
	(0.033)	(0.041)	(0.052)	(0.037)	(0.045)	(0.059)
Negative years	18/18	17/17	15/15	18/18	17/17	15/15
Observations	49,278	41,237	29,025	242,952	203,308	143,100
BDCs	192	167	107	192	167	107

Notes: Each column regresses the change in an industry's share of a BDC's debt principal, from its average over a baseline year to the fourth quarter one, two or four years later, on the baseline share, with BDC-by-year and industry-by-year fixed effects. Shares are in percentage points. Baseline years are 2007–2024, 2007–2023 and 2007–2021, respectively. Standard errors, in parentheses, are two-way clustered by BDC and industry; BDC-bootstrap p-values are below 0.001 in every column. The decline comes from borrower exits, partly offset by new borrowers. "Negative years" counts the baseline years in which a separate cross-sectional regression gives a negative coefficient.